

FEATURE-SELECTION RISK

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ABSTRACT. Companies have overlapping exposures to many different features that might plausibly affect their returns, like whether they're involved in a crowded trade, whether they're mentioned in an M&A rumor, or whether their supplier recently missed an earnings forecast. Yet, at any point in time, only a handful of these features actually matter. As a result, traders have to simultaneously infer both the identity and the value of the few relevant features. I show theoretically that, when traders face this sort of joint inference problem, the risk of selecting the wrong features can spill over and distort how they value assets—that is, the high-dimensional nature of modern financial markets can act like a cognitive constraint even if traders themselves are fully rational. Moreover, I show how modeling feature-selection risk leads to additional predictions that are outside the scope of noise-trader risk. For instance, to discover pricing errors as quickly as possible, a model with feature-selection risk suggests that traders should simultaneously trade a random assortment of complex, heterogeneous assets rather than Arrow securities. Empirically, I find that using an estimation strategy that explicitly accounts for traders' joint inference problem increases out-of-sample return predictability at the monthly horizon by 144.3%, from $R^2 = 3.65\%$ to 9.35%, suggesting that this feature-selection problem is important to real-world traders.

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Current Version: <http://www.alexchinco.com/feature-selection-risk.pdf>.

1. INTRODUCTION

In an efficient market, if a few stocks suddenly get mispriced because they share a common feature, like being involved in a crowded trade or getting mentioned in an M&A rumor, then fully-rational traders should rapidly exploit and eliminate this error. However, markets don't always appear efficient, and people have suggested a variety of trader shortcomings to explain why. For instance, traders might face limits to arbitrage as in [Miller \(1977\)](#), suffer from cognitive biases as in [Daniel, Hirshleifer, and Subrahmanyam \(1998\)](#), or exhibit the symptoms of non-standard preferences as in [Barberis, Huang, and Santos \(2001\)](#).

But, is it always right to blame traders? Perhaps a market's inefficiency has more to do with its dimensions than with its traders' limitations? After all, modern financial markets are extremely complex and densely interconnected. For any given pricing error, there are often many plausible explanations. Did Callaway Golf's stock just plunge because it happened to be involved in a crowded short-term trading strategy? Or, was it because there's some truth to that new rumor about Callaway acquiring Fortune Brand? A trader should respond differently to each of these hypotheses, shorting the other stocks in the crowded strategy in the first case and buying shares of Fortune in the second. In a high-dimensional setting where assets can share many overlapping features, that is, where Callaway can be both involved in a crowded trade and mentioned in an M&A rumor, markets don't always provide enough information to sort through the many competing hypotheses.

This paper shows that, when traders have to simultaneously decide both which features are mispriced and how they should be correctly valued, the risk of selecting the wrong features can spill over and distort how they value assets. The high-dimensional nature of modern financial markets can act like a cognitive constraint even if traders are fully rational.

Illustrative Example. Let's take a look at a short example to see how. Imagine you're a trader, and each company's stock returns can have exposure to any combination of 7 features: 1) whether the company's involved in a crowded trade ([Khandani and Lo \(2007\)](#)), 2) whether it's been mentioned in a news article about M&A activity ([D'Aspremont and Luss \(2012\)](#)), 3) whether there's been an announcement about its major supplier ([Cohen and Frazzini \(2008\)](#)), 4) whether its labor force has recently unionized ([Klasa, Maxwell, and Ortiz-Molina \(2009\)](#)), 5) whether it belongs to the tobacco industry ([Hong and Kacperczyk \(2009\)](#)), 6) whether it's been referenced in a scientific journal article ([Huberman and Regev \(2001\)](#)), and 7) whether it's been included in the S&P 500 ([Barberis, Shleifer, and Wurgler \(2005\)](#)). Moreover, suppose you have a hunch that there's been a shock to one of these features, but you don't initially know which one. All you know is that the market hasn't fully appreciated the shock, and stocks with this mystery feature will realize abnormal returns of $\alpha > 0$ in the near future. Here is the question: How many stock returns do you need to see in order to figure out which, if any, of the shocks has occurred?

Answer: 3.

Suppose the first company has exposure to features $\{1,3,5,7\}$ —that is, it's involved in a crowded trade, there's been an announcement about its major supplier, it belongs to the tobacco industry, and it's been recently added to the S&P 500. Similarly, suppose that the second company has exposure to features $\{2,3,6,7\}$ and the third company has exposure to features $\{4,5,6,7\}$. The abnormal returns for these three stocks always reveals exactly which feature-specific shock has occurred. For instance, if only the first stock has positive abnormal returns, $ar_1 = \alpha$ while $ar_2 = ar_3 = 0$, then it must have been a crowded-trade-specific shock:

$$\begin{bmatrix} ar_1 \\ ar_2 \\ ar_3 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (1)$$

Whereas, if both the first and the third stock have positive abnormal returns, $ar_1 = ar_3 = \alpha$ while $ar_2 = 0$, then it must have been a shock to the tobacco industry. Crucially, there is no way to identify which feature-specific shock has occurred using fewer stocks. At the end of the day, you need at least 8 bits of information to pick out which of the 7 feature-specific shocks has occurred and rule out the possibility of no change and $2^3 = 8$.

Now, to see how this fact relates to market efficiency, let's rewind the clock a bit and consider the problem you face after seeing only the first two companies' abnormal returns, $ar_1 = \alpha$ and $ar_2 = 0$:

$$\begin{bmatrix} \alpha \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}. \quad (2)$$

Since the first and the fifth columns are $[1 \ 0]^\top$, you know that either a crowded-trade-specific shock has occurred or a tobacco-industry-specific shock has occurred. It has to be one of these two features since these are the only two features that the first company has exposure to but the second company doesn't. So, what's the right way to value the third company's stock which has exposure to features $\{4,5,6,7\}$, meaning that it's in the tobacco industry but not involved in a crowded trade?

There are two possibilities. If the crowded-trade-specific shock has occurred, then you should leave the third company's value unchanged; whereas, if the tobacco-industry-specific shock has occurred, then you should revise your valuation. Thus, after seeing only two observations, you have to split the difference. If it was in fact the tobacco-industry-specific shock, then you will only update half-way, and it will look like you were slow to react to public information. By contrast, if it was the crowded-trade-specific shock, then, when you revise your valuation of the third company's stock half-way, it will look like you were

over-reacting to noise. Nevertheless, this is the best you can do in real time. It's not like you're making some cognitive error or fighting against some trading friction. Instead, it's the dimensionality of your inference problem that's generating the extra risk, that's warping your perception of the third company's value, that's distorting prices.

Feature-Selection Bound. Of course, this is just a stylized example. There are only a handful of assets, each asset's feature exposures are hand-picked, and their fundamental values don't reflect the standard risk factors. To address these concerns, I apply tools from the compressed-sensing literature to generalize the result and show that if traders have seen fewer than $N^*(Q, K)$ observations,¹

$$N^*(Q, K) \asymp K \cdot \log(Q/K), \quad (3)$$

then, no matter what inference strategy they use, traders cannot always identify which features have realized a shock in a large market with an arbitrary number of features, Q , and an arbitrary number of shocks, K . This feature-selection bound holds even when feature exposures are randomly assigned and when companies' fundamental values reflect the usual risk factors. Moreover, in the presence of noise, some feature-selection risk will remain even after the bound has been reached. Thus, feature-selection risk is endemic to any high-dimensional market. If assets share many overlapping features, then markets might not provide enough information to pinpoint exactly which ones matter. It's as if the high-dimensional nature of the market is introducing a cognitive constraint even though the traders themselves are fully rational.

Asset-Pricing Model. In order to quantify the extent to which feature-selection risk limits market efficiency, I study a Kyle (1985)-type model with N assets whose values are a function of $K \ll Q$ feature-specific shocks. The model is entirely standard except for one key detail: uninformed traders, like the market maker and any would-be arbitrageurs, don't know ahead of time which K feature-specific shocks to analyze. For instance, if there is a lot of demand for Callaway Golf, the market maker now has to ask herself: Is this demand saying something about a feature-specific shock? If so, which one? Or, did Callaway just happen to realize a large noise-trader demand shock? By adding this simple twist, it's possible to extend the standard information-based asset-pricing models to allow for feature-selection risk. When uninformed traders now have to infer both the identity and the size of the $K \ll Q$ feature-specific shocks, they are less responsive to aggregate demand shocks than in the original Kyle (1985) setup, making equilibrium prices less accurate.

Just Noise-Trader Risk? Next, I discuss how feature-selection risk differs from noise-trader risk. To start with, I show mathematically that feature-selection risk and noise-trader risk are substitutes. If the market maker has to sort through a sufficiently large number of

¹ $f_N \asymp g_N$ denotes asymptotically bounded above and below, implying both $f_N = O(g_N)$ and $g_N = O(f_N)$.

potentially relevant features, then she is likely to make a feature-selection error, regardless of the volatility of noise-trader demand. What's more, real-world traders often try to exploit this fact. For example, the co-CEO of Renaissance Technologies, Robert Mercer, has pointed out that "some signals that make no intuitive sense do indeed work... The signals that we have been trading without interruption for 15 years make no sense, otherwise someone else would have found them."² Such signals are hidden in a large feature space rather than behind a lot of noise-trader demand volatility.

Then, I show that feature-selection risk makes novel predictions that are outside the scope of noise-trader risk. For instance, the textbook approach to learning suggests using one Arrow security for each risk. Yet, the compressed-sensing theory used in the current paper says that a trader can identify feature-specific shocks using far fewer assets if the assets are extremely complex and heterogeneous. So, to identify feature-specific shocks as soon as possible, sophisticated traders should trade a random assortment of complex, heterogeneous assets rather than simple stocks or bonds.

Empirical Evidence. Finally, I give empirical evidence that real-world traders actually care about solving this joint inference problem. To do this, I collect data on 79 different monthly factors that have been used in the asset-pricing literature. Then, for each NYSE stock I analyze 24-month rolling windows from January 1990 to December 2010, estimate the $K \ll 79$ important factor loadings using a penalized-regression procedure, and predict the stock's excess return in the subsequent month. This penalized regression sets all of the "smaller" factor loadings to zero and makes it possible to estimate a sparse subset of the 79 coefficients using only 24 months of data, an approach that would clearly be unidentified using the standard regression techniques. Using this estimation strategy that explicitly accounts for traders' feature-selection problem increases the accuracy of out-of-sample return predictions at the monthly horizon by 144.3%, from $R^2 = 3.65\%$ to $R^2 = 9.35\%$! Thus, solving this joint inference problem is very important for real-world traders.

2. BASELINE EQUILIBRIUM MODEL

Let's begin by characterizing a baseline equilibrium where traders don't face any feature-selection risk. Specifically, I assume that they have access to an oracle that alerts them to the K features that have realized a shock, but not the size or sign of the shock. We can then return to this model during the later analysis as a point of comparison to answer the question: how does feature-selection risk alter the usual predictions?

²Mallaby, S. (2010) *More Money Than God* (1 ed.) Penguin Books.

2.1. Market Structure. I study a static market with N assets whose fundamental values, v_n , are governed by their exposure to Q features,

$$v_n = \sum_{q=1}^Q \alpha_q \cdot x_{n,q}, \quad (4)$$

where $x_{n,q} \stackrel{\text{iid}}{\sim} N(0,1)$ denotes asset n 's exposure to the q th feature and α_q denotes the size of the shock to feature q . So, for example, if there is a shock of size $\alpha_{\text{Tobacco}} = \1 to stocks in the tobacco industry, then the share price of a company in that industry, $x_{n,\text{Tobacco}} = 1$, will rise by \$1.

I consider the setting where everyone knows each asset's feature exposures $\mathbf{x}_n$. That is, all agents have a detailed list of whether or not each asset has been involved in a crowded trade, mentioned in an article on M&A activity, suffered a setback to one of its suppliers, etc. . . If there is any uncertainty in later sections, it will be about which elements in $\boldsymbol{\alpha}$ are non-zero. For instance, traders might be uncertain about whether or not the tobacco industry has realized a shock, but they will never be uncertain about whether a particular company is in the tobacco industry.

Heterogeneous Exposures. Because each asset has different feature exposures, each asset will manifest a feature-specific shock in a slightly different way. For example, we know that some stocks are more likely to be included in statistical arbitrage strategies than others, news of M&A activity has an opposite effect on the acquirer and the target, and some companies are more strongly impacted by news about a particular supplier than others. Suppose that asset 1 has exposures to the stat-arb-strategy, M&A activity, and supplier stock features given by $\mathbf{x}_1 = [1.50 \ 0.50 \ -0.10]^\top$ while asset 2 has feature exposures $\mathbf{x}_2 = [-0.50 \ -0.75 \ 1.00]^\top$. Each stock's value will then be:

$$v_1 = \alpha_{\text{StatArb}} \times (+1.50) + \alpha_{\text{M\&A}} \times (+0.50) + \alpha_{\text{EconLink}} \times (-0.10) + \dots \quad (5a)$$

$$v_2 = \alpha_{\text{StatArb}} \times (-0.50) + \alpha_{\text{M\&A}} \times (-0.75) + \alpha_{\text{EconLink}} \times (+1.00) + \dots \quad (5b)$$

Thus, a positive M&A activity shock of $\alpha_{\text{M\&A}} = 1$ will lead to a \$0.50 rise in the fundamental value of asset 1. By contrast, the same shock will lead to a \$0.75 decline in the fundamental value of asset 2. Same shock. Different feature exposures. Opposite affects on value.

Sparse Shocks. To capture the idea that only a few of the many possible features that might impact a stock's value each period actually matter, I study a world where only K of the elements in $\boldsymbol{\alpha}$ are non-zero,

$$K = \|\boldsymbol{\alpha}\|_0 = \sum_{q=1}^Q 1_{\{\alpha_q \neq 0\}}, \quad (6)$$

with $Q \gg N \geq K$ and $\mathcal{K}$ denoting the set of shocked features. I also assume the vector of feature-specific shocks, $\boldsymbol{\alpha}$, satisfies the following conditions:

- (1) $\mathcal{K} \subset \{1, 2, \dots, Q\}$ is selected uniformly at random.

- (2) The signs of $\alpha_{[\mathcal{K}]}$ are independent and equally likely to be -1 or $+1$.
- (3) The magnitudes of $\alpha_{[\mathcal{K}]}$ are independent and bounded by $\alpha_{\max} \geq |\alpha_q| > \sigma_z$.

To be sure, shocks are never really exactly sparse; they are only approximately sparse meaning that they may be well approximated by sparse expansions. All of the results in this paper go through if you assume that K features realize shocks that are much bigger than the rest.

Feature Selection. This market structure means that it's possible for a trader to see several assets behaving wildly without being able to put his finger on which K feature-specific shocks are the culprit. For instance, the chairman of Caxton Management, Bruce Kovner, notes that there are often many plausible reasons why prices might move in either direction at any point in time. "During the past six months, I had good arguments for the Canadian dollar going down, and good arguments for the Canadian dollar going up. It was unclear to me which interpretation was correct."³ This wasn't a situation where Kovner had to learn more about a well-defined trading opportunity; rather, the challenge was to pick which explanation to trade on in the first place. Kovner faced feature-selection risk.

Of course, sometimes traders aren't in the business of spotting feature-specific shocks. For example, a January 2008 Chicago Tribune article about Priceline.com reported that "a third-quarter earnings surprise sent [the company's] shares skyward in November, following an earlier announcement that the online travel agency planned to make permanent a no-booking-fees promotion on its airline ticket purchases."⁴ No one was confused about why Priceline's price rose. The only problem facing traders was deciding how much to adjust the price. Existing information-based asset-pricing models are well suited to this setting.

2.2. Objective Functions. There are two kinds of optimizing agents, asset-specific informed traders and a market-wide market maker, along with a collection of asset-specific noise traders.

Informed Traders' Problem. Asset-specific informed traders know the fundamental value of a single asset, v_n , and solve the standard static Kyle (1985)-type optimization problem with risk neutral preferences,

$$\max_{y_n \in \mathbf{R}} \mathbf{E}[(v_n - p_n) \cdot y_n \mid v_n], \quad (7)$$

where y_n denotes the size of asset n 's informed trader's market order in units of shares and p_n denotes the price that they pay in units of dollars per share. Crucially, for these traders, the fundamental value of each asset is just a random variable with no further structure. They do not observe which K feature-specific shocks govern its value.

There are a couple of ways to justify this assumption. First, you might think about the asset-specific informed traders as value investors. For instance, Li Lu, founder of Himalaya

³Schwager, J. (1989) *Market Wizards: Interviews with Top Traders*. (1 ed.) New York Institute of Finance.

⁴DiColo, J. (2008, Jan. 20) Priceline's Power Looks Promising in Europe, Asia. *Chicago Tribune*.

Capital and well known value investor, suggests that in order to gain market insight you should “Pick one business. Any business. And truly understand it. I tell my interns to work through this exercise—imagine a distant relative passes away and you find out that you have inherited 100% of a business they owned. What are you going to do about it?”⁵ It’s like they have an informative gut instinct. Alternatively, you can think about the informed traders as getting a signal about the level of noise-trader demand in a given asset. They can then invert this information about noise-trader demand to learn something about the asset’s fundamental value, v_n , without learning anything about its structure.

Market Maker’s Problem. The market-wide market maker observes aggregate order flow, d_n , for each of the N assets,

$$d_n = y_n + z_n, \quad (8)$$

which is composed of demand from the asset-specific informed traders, y_n , and from asset-specific noise traders, $z_n \stackrel{\text{iid}}{\sim} N(0, \sigma_z^2)$. He then tries to set the price of each asset as close as possible to its fundamental value given the cross-section of aggregate demand:

$$\min_{\mathbf{p} \in \mathbf{R}^N} E \left(\frac{1}{N} \cdot \sum_{n=1}^N (p_n - v_n)^2 \mid \mathbf{d} \right). \quad (9)$$

Put differently, competitive pressures force the market maker to try and minimize the mean squared error between the price and each asset’s value.

Notice that this formulation of the market maker’s problem is slightly different from the one in the original Kyle (1985) model. Here, the market maker explicitly minimizes his prediction error; whereas, in the original setup, the market maker just sets the price equal to his conditional expectation, which happens to minimize his prediction error since there are as many assets as shocks. In the current paper, it’s important that the market maker explicitly minimizes his prediction error because the conditional expectation will no longer be well defined when there are more possible feature-specific shocks than assets.

Because there are many more features than assets, $Q \gg N \geq K$, the market maker must use a feature-selection rule $\phi(\mathbf{d}, \mathbf{X})$ that accepts an $(N \times 1)$ -dimensional vector of aggregate demand as well as an $(N \times Q)$ -dimensional matrix of features and then returns a $(Q \times 1)$ -dimensional vector of estimated feature-specific shocks:

$$\phi : \mathbf{R}^N \times \mathbf{R}^{N \times Q} \mapsto \mathbf{R}^Q. \quad (10)$$

I use $\hat{\boldsymbol{\alpha}} = \phi(\mathbf{d}, \mathbf{X})$ to denote the estimated shocks. Later, I will give bounds on how well the best possible feature-selection rule can perform in a market with Q features, K shocks, and N assets. The nature of the equilibrium asset prices will depend on how much information about the sparse feature-specific shocks, $\boldsymbol{\alpha}$, the market maker can tease out of the cross-section of aggregate demand, $\mathbf{d}$. It’s clear that real-world traders worry about how much

⁵Lu, L. (2010) *Lecture at Columbia Business School*.

Timing in Oracle Equilibrium

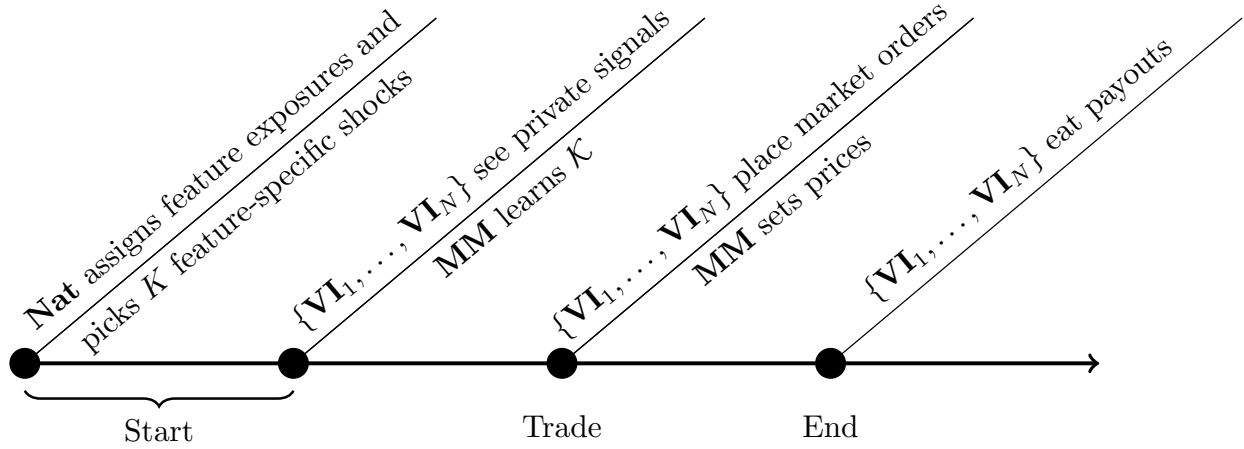


Figure 1. What each agent knows and when they know it in the model where the common market maker knows which K features have realized a shock.

their market maker can learn from the *combination* of their orders. For instance, quantitative hedge funds place the orders for different legs of the same trade with different brokers to make it difficult for their brokers to do exactly this sort of reverse engineering.

2.3. Oracle Equilibrium. Let's now explore the equilibrium when the market maker has an oracle telling him exactly which K features have realized a shock. It turns out that the coefficients in Proposition 2.3 below are identical to the standard Kyle (1985) model coefficients. This fact highlights how existing information-based asset-pricing models implicitly assume that all traders know exactly which features to study.

Model Timing. Figure 1 summarizes the timing of the model. First, nature assigns feature exposures to the N assets and picks a subset of K features to realize shocks. After the exposures and shocks have been drawn but before any trading takes place, the N asset-specific informed traders learn the fundamental value of their own asset, v_n , and the single market maker common to all N assets observes which K features have realized a shock (but not the size or sign of these shocks). Finally, trading takes place. Each of the N informed traders and noise traders places a market order. Then, the market maker observes each asset's aggregate order flow, updates his conditional expectation about their values, and sets prices accordingly.

Equilibrium Definition. An equilibrium, $\mathcal{E} = \{\theta, \lambda\}$, is a linear demand rule for each of the N asset-specific informed traders,

$$y_n = \theta \cdot v_n, \quad (11)$$

and a linear pricing rule for the single market maker common to all N assets,

$$p_n = \lambda \cdot d_n, \quad (12)$$

such that a) the demand rule θ solves Equation (7) given the correct assumption about λ and b) the pricing rule λ solves Equation (9) given the correct assumption about θ .

Proposition 2.3 (Oracle Equilibrium). *If the market maker knows $\mathcal{K}$, then there exists an equilibrium defined by coefficients:*

$$\lambda = \frac{1}{2 \cdot \theta} \quad (13a)$$

$$\theta = \sqrt{\frac{K}{N}} \cdot \left(\frac{\sigma_z}{\sigma_v} \right). \quad (13b)$$

Equilibrium Characterization. Because there are more assets than feature-specific shocks, the market maker can just run the standard OLS regression,

$$1/\theta \cdot d_n = \mathbf{x}_n \hat{\boldsymbol{\alpha}}_{\text{OLS}} + \epsilon_n, \quad (14)$$

to estimate $\hat{\boldsymbol{\alpha}}_{\text{OLS}}$. Knowing these coefficients then gives him an unbiased signal, $\mathbf{X} \hat{\boldsymbol{\alpha}}_{\text{OLS}}$, about each asset's fundamental value. This signal has variance

$$\mathbb{E} \left[\frac{1}{N} \cdot \|\mathbf{v} - \mathbf{X} \hat{\boldsymbol{\alpha}}_{\text{OLS}}\|_2^2 \right] = \frac{K}{N} \cdot \frac{\sigma_z^2}{\theta^2}, \quad (15)$$

where $\mathbf{v}$ is an $(N \times 1)$ -dimensional vector of asset values. Using his priors on the distribution of each asset's value, $v_n \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma_v^2)$, he can then use DeGroot (1969) updating to form posterior beliefs. The market maker's signal error is increasing in the variance of noise-trader demand, so he has a harder time figuring out if a positive demand shock is due to noise traders or just a really strong fundamental value realization. Thus, more noise trader demand volatility means informed traders have an easier time masking their trades allowing them to trade more intensely.

3. FEATURE-SELECTION BOUND

We just saw what the equilibrium looks like when traders know exactly which features to analyze. Let's now look at how hard it is to recover this information without an oracle. Specifically, I show that, if the market maker hasn't seen at least $N^*(Q, K)$ observations, then he will always suffer from feature-selection risk and will always make some errors in picking which features to analyze, no matter what inference strategy he uses and even if he is fully rational.

3.1. Theoretical Minimum. Suppose that the market maker was the most sophisticated trader ever and could choose the best inference strategy possible, ϕ_{Best} . How many observations does he need to see to be sure that he's identified which feature-specific shocks have

taken place? He doesn't need to see Q observations since the vector α is K -sparse. But what is this bare minimum number?

Large-Market Asymptotics. To answer this question, I consider limiting results for sequences of markets $\{(Q_N, K_N)\}_{N \geq 0}$ where the number of features, $Q = Q_N$, and the sparsity level, $K = K_N$, are allowed to grow with the number of observations, N :

$$\lim_{N \rightarrow \infty} Q_N, K_N = \infty \quad N \geq K_N \quad \lim_{N \rightarrow \infty} K_N/Q_N = 0. \quad (16)$$

For example, take $K = \sqrt{Q}$. This asymptotic formulation captures the spirit of traders' joint inference problem. For instance, Daniel (2009) notes that during the Quant Meltdown of August 2007 "markets appeared calm to non-quantitative investors. . . you could not tell that anything was happening without quant goggles" even though large funds like Highbridge Capital Management were suffering losses on the order of 16%.⁶ All stocks with exposure to the held-in-a-stat-arb-strategy feature realized a massive shock, but this feature was just one of many plausible feature-specific shocks that might have occurred ex ante. Unless you knew where to look (had "quant goggles"), the event just looked like noise.

Feature-Selection Error. I define the market maker's feature-selection error as the quantity

$$\text{FSE}[\phi] = \mathbb{E} [\|S[\hat{\alpha}] - S[\alpha]\|_\infty] \quad (17)$$

where the operator $S[\cdot]$ identifies the support of a vector:

$$S[\hat{\alpha}_q] = \begin{cases} 1 & \text{if } \hat{\alpha}_q \neq 0 \\ 0 & \text{if } \hat{\alpha}_q = 0 \end{cases}. \quad (18)$$

The ℓ_∞ -norm gives a 1 if there is any difference in the support of the vectors and a 0 otherwise. In words, $\text{FSE}[\phi]$ is the probability that the market maker's selection rule, ϕ , chooses the wrong subset of features when averaging over not only the measurement noise but also the choice of the Gaussian exposure matrix, $\mathbf{X}$. Let Φ denote the set of all possible inference strategies the market maker might use. If there exists some inference strategy $\phi \in \Phi$ with $\text{FSE}[\phi] = 0$, then the market maker can use this approach to always select which feature-specific shocks have taken place with probability 1. That is to say, there exists (at least in principle) an inference strategy that would be just as good as having an oracle. It may not be computationally feasible, but it would exist.

Feature-Selection Bound. The feature-selection bound given in Proposition 3.1 below then says that no such strategy exists when the market maker has seen fewer than $N^*(Q, K)$ observations. When $N < N^*(Q, K)$, at least a few feature-selection errors are unavoidable regardless of what approach $\phi \in \Phi$ the market maker takes.

⁶Zuckerman, G., J. Hagerty, and D. Gauthier-Villars (2007) Mortgage Crisis Spreads. *Wall Street Journal*.

Proposition 3.1 (Feature-Selection Bound). *If there exists some constant $C > 0$ such that*

$$N < C \cdot K_N \cdot \log(Q_N/K_N) \quad (19)$$

as $N \rightarrow \infty$, then there exists some constant $c > 0$ such that

$$\min_{\phi \in \Phi} \text{FSE}[\phi] > c. \quad (20)$$

The threshold value $N^(Q, K) \asymp K \cdot \log(Q/K)$ is the feature-selection bound.*

Importantly, Proposition 3.1 doesn't make any assumptions about the market maker's cognitive abilities. It says that when $N < N^*(Q, K)$ the market maker has to be misinterpreting aggregate-demand signals at least some of the time due to the nature of his sparse, high-dimensional, inference problem. Put another way, this minimum number of observations is a consequence of a theoretical bound on how informative market signals can be rather than a consequence of thinking costs or trading frictions. In some sense, it has nothing to do with the market maker. He could be Einstein, Friedman, and Kasparov all rolled into one and it wouldn't matter. There is simply a lower bound on the amount of data needed to say anything useful about which market events have taken place using the cross-section of aggregate demand. This is a very different way of thinking about why rational traders sometimes misinterpret market signals. This result is first derived in [Wainwright \(2009a\)](#).

3.2. Discussion. There are a couple of points about the interpretation of Proposition 3.1 worth discussing in more detail.

Choice of Asymptotics. First, while the asymptotics are helpful for analytical reasons, they are not critical to the underlying result. There is a qualitative change in the nature of any inference problem when you move from choosing which feature-specific shocks have occurred to deciding how large they must have been. To see why, let's return to the example in Section 1 where only 1 of the 7 features might have realized a shock, and consider the more general case where any of the 7 features could have. This gives

$$\begin{aligned} 128 &= \binom{7}{0} + \binom{7}{1} + \binom{7}{2} + \binom{7}{3} + \binom{7}{4} + \binom{7}{5} + \binom{7}{6} + \binom{7}{7} \\ &= 1 + 7 + 21 + 35 + 35 + 21 + 7 + 1 \\ &= 2^7 \end{aligned} \quad (21)$$

different feature combinations. Thus, $N^*(7, 7) = 7$ provides a trader just enough differences to identify which combination of features has realized a shock. More generally, for any number of features, Q , a trader needs $2^Q = \sum_{k=0}^Q \binom{Q}{k}$ observations to detect shocks if he has no information about K . This gives an information theoretic interpretation to the meaning of "just identified" that has nothing to do with linear algebra or matrix invertibility.

Applying the Bound. Second, these asymptotics do not pose a practical problem when

applying the bound. To begin with, real-world markets are finite but very large, so the asymptotic approximation is a good one. While it isn't possible to give a precise formulation of the feature-selection bound in the finite sample case, practical compressed sensing techniques can make error-rate guarantees in finite samples. What's more, analysts regularly make this sort of asymptotic-to-finite leap in mainstream econometric applications. For example, the practical implementation of GMM involves a 2-step procedure as outlined in [Newey and McFadden \(1994\)](#). The first step estimates the coefficient vector using the identity weighting matrix on the basis that any positive-semidefinite weighting matrix will give the same point estimates in the large T limit. The second step then uses the realized point estimates to compute the coefficient standard errors.

Best-Case Result. Finally, the result in Proposition 3.1 is likely too optimistic about the ability of the most sophisticated market maker since it makes no assumptions about the inference strategy being convex. How much harder could the non-convex approach be? A lot. Consider the motivating example from Section 1. Suppose that $Q = 400$, and I told you that exactly $K = 5$ of the features were mispriced. You can certainly try to solve the general problem by tackling each of the $\binom{400}{5} \approx 8.3 \times 10^{10}$ sub-problems with a regression procedure; however, this is a huge number of cases to check on par with the number of bits in the human genome. As [Rockafellar \(1993\)](#) writes, "the great watershed in optimization isn't between linearity and nonlinearity, but convexity and non-convexity."

4. FEATURE-SELECTION RISK

If the market maker hasn't seen at least $N^*(Q, K)$ observations, then he will always suffer from feature-selection risk no matter what inference strategy he uses. Let's now introduce this feature-selection risk into the baseline asset-pricing model to see how it warps the market maker's perception of asset values. The basic equilibrium concept will remain completely standard. The key question to ask is: how much information about which K feature-specific shocks have occurred can the market maker infer from the cross-section of aggregate demand of N stocks?

4.1. Inference Strategy. In order to solve for equilibrium asset prices, I need to be able to compute the market maker's posterior beliefs after observing the cross-section of aggregate demand, $\mathbf{d}$. This means that I need to make a choice about which inference strategy the market maker uses.

Using the LASSO. I study a market maker who uses the least absolute shrinkage and selection operator (LASSO) outlined in [Tibshirani \(1996\)](#),

$$\hat{\boldsymbol{\alpha}} = \arg \min_{\boldsymbol{\alpha} \in \mathbf{R}^Q} \left\{ \|\mathbf{X}\tilde{\boldsymbol{\alpha}} - (1/\theta) \cdot \mathbf{d}\|_2^2 + \gamma \cdot \|\tilde{\boldsymbol{\alpha}}\|_{\ell_1} \right\}, \quad (22)$$

for $\gamma > 0$. The ℓ_1 norm means that the LASSO sets all coefficient estimates with $|\alpha_q| < \gamma$

equal to zero. It generates a preference for sparsity. For example, if there were no $\gamma \cdot \|\tilde{\alpha}\|_{\ell_1}$ term, then the inference strategy would be equivalent to ordinary least squares which isn't well-posed for $Q \gg N$. The tuning parameter, γ , controls how likely the estimation procedure is to get false positives. To screen out spurious variables, you want γ to be large; however, increasing γ also means that you are more likely to ignore meaningful variables that happen to look small in the data by chance. Decreasing γ to reduce this problem floods the results with spurious coefficients.

Why the ℓ_1 -Norm? Note that in the current paper, the use of the ℓ_1 -norm is not a consequence of bounded rationality as in [Gabaix \(2013\)](#). Rather, it is simply a way for the market maker to draw an inference about the value of each asset given the cross-section of aggregate demand. Since the market maker doesn't have access to an oracle, there are now more features than stocks, $Q \gg N$. As a result, his inference procedure needs to have a preference for sparsity. Any penalty with a norm $p \in [0,1]$ will do. For example, think about the ℓ_0 problem:

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^Q} \{ \|\mathbf{X}\alpha - (1/\theta) \cdot \mathbf{d}\|_2^2 + \gamma \cdot \|\alpha\|_{\ell_0} \}. \quad (23)$$

However, a penalty with a norm $p \in [0,1)$ generates a non-convex inference problem which is computationally intractable. [Natarajan \(1995\)](#) explicitly shows that ℓ_0 constrained programming is NP-hard. Thus, the ℓ_1 norm, which sits right on the boundary of the two regions, is the natural choice for the penalty. What's more, when feature exposures are drawn independently from identical Gaussian distributions as they are in the current paper, the LASSO comes within a logarithmic factor of optimality as shown in [Wainwright \(2009b\)](#).

4.2. Equilibrium Using the LASSO. We can now solve for the equilibrium coefficients in the more general setting where the market maker doesn't have access to an oracle and must solve a sparse, high-dimensional, inference problem on his own. I show that informed traders in this new model earn higher profits since they can hide behind both noise-trader demand volatility and feature-selection risk.

LASSO Error Rate. [Candes and Plan \(2009\)](#) prove that if the market maker sees the aggregate demand for at least $N^*(Q,K)$ assets, then the LASSO gives a signal about each asset's value, $\mathbf{v}$, with a signal error that satisfies the inequality below,

$$\frac{1}{N} \cdot \|\mathbf{X}\hat{\alpha}_{\text{LASSO}} - \mathbf{v}\|_2^2 \leq \tilde{C}^2 \cdot \log(Q) \times \frac{K}{N} \cdot \frac{\sigma_z^2}{\theta^2}, \quad (24)$$

with probability approaching unity as $N \rightarrow \infty$ for $\tilde{C} = 2 \cdot \sqrt{2} \cdot (1 + \sqrt{2})$. Where does this $\tilde{C}^2 \cdot \log(Q)$ factor come from? Because the market maker has to simultaneously decide both *which* asset features have realized a shock and also *how large* they were, he will sometimes make errors in identifying which features have realized a shock. When he does so, there will be additional noise in his posterior beliefs about each asset's fundamental value. It's

these feature-selection errors that increase the variance of his posterior beliefs by a factor $\tilde{C}^2 \cdot \log(Q)$ relative to when he had an oracle.

Equilibrium Definition. The equilibrium concept will be the same as before. An equilibrium, $\mathcal{E}_\phi = \{\theta, \lambda\}$, is a linear demand rule for each of the N asset-specific informed traders,

$$y_n = \theta \cdot v_n, \quad (25)$$

and a linear pricing rule for the single market maker common to all N assets,

$$p_n = \lambda \cdot d_n, \quad (26)$$

such that a) the demand rule θ solves Equation (7) given the correct assumption about λ and b) the pricing rule λ solves Equation (9) given the correct assumption about θ and assuming the market maker uses the LASSO to solve his sparse, high-dimensional, inference problem.

Proposition 4.2 (Equilibrium Using the LASSO). *If the market maker uses the LASSO with $\gamma = 2 \cdot (\sigma_z/\theta) \cdot \sqrt{2 \cdot \log(Q)}$ to identify and interpret feature-specific shocks and $N > N^*(Q, K)$, then there exists an equilibrium defined by coefficients*

$$\lambda = \frac{1}{2 \cdot \theta} \quad (27a)$$

$$\theta = C \cdot \sqrt{\log(Q)} \times \sqrt{\frac{K}{N}} \cdot \left(\frac{\sigma_z}{\sigma_v} \right) \quad (27b)$$

for some positive numerical constant $0 < C < \tilde{C}$.

Equilibrium Characterization. The key takeaway from Proposition 4.2 is that increasing Q , the number of payout-relevant features that a market maker has to sort through, makes the price less responsive to demand shocks. This happens via two different channels. First, increasing Q raises the feature-selection bound, $N^*(Q, K)$, so that the market maker has to see more assets before he can correctly identify which features have realized a shock. When there are fewer than $N^*(Q, K)$ assets for the market maker to inspect, the LASSO doesn't reveal anything about which feature-specific shocks have occurred. Thus, in this regime, the common market maker effectively operates in N distinct asset markets. Each asset's demand gives him information about that particular asset's fundamental value, but he can't extrapolate this information from one asset to the next. Second, increasing Q makes the market maker less certain about his inferences. It imposes a penalty on the precision of the market maker's posterior beliefs of $C^2 \cdot \log(Q)$ per unit of fundamental volatility. In short, it takes time to decode market signals.

Numerical Constant. Proposition 4.2 includes a numerical constant C . The exact value of this numerical constant will depend on the distribution of the sizes of the K feature-specific shocks. The exact value of the constant can be found numerically by bootstrap procedures—that is, by repeatedly estimating the LASSO on sample datasets. For example, when the

magnitude of the K feature-specific shocks is drawn $\alpha_q \sim^{\text{iid}} \pm \text{Unif}[1,2] \cdot (\sigma_z/\theta)$, simulations reveal that $C \approx 2 \cdot (1 + \sqrt{2}) \approx 4.82$. I make no effort to characterize this value further because it depends on the gritty details of the asset value distribution. Changing C slightly does not alter the qualitative intuition behind the impact of feature-selection risk.

5. JUST NOISE-TRADER RISK?

Increasing the number of potentially relevant features, Q , that the market maker has to sort through makes the equilibrium price less responsive to aggregate demand shocks. This sounds similar to the effect of increasing noise-trader demand volatility. Is feature-selection risk just noise-trader risk in disguise? No. It is not. Let's now turn our attention to how feature-selection risk differs from noise-trader risk.

5.1. Substituting Risks. First, I show that feature-selection risk and noise-trader risk are substitutes. The model outlined above predicts that it should be more profitable for an informed trader to learn a firm-specific piece of news in a markets with a larger number of payout-relevant features—that is, with a larger value of Q —regardless of the level of noise-trader risk.

Informed-Trader Profit. Forcing market makers to sort through a larger number of potentially relevant features makes them less responsive to informed trader demand. To make this point precise, consider the unconditional expectation of an informed trader's profits when the market maker has to use the LASSO to both identify and value feature-specific shocks:

$$\Pi(Q, \sigma_z) = \mathbb{E} \left(\max_{y_n} \mathbb{E} [(v_n - p_n) \cdot y_n \mid v_n] \right). \quad (28)$$

Proposition 6.1 below shows that this quantity is increasing in the number of payout-relevant features. Put differently, a bigger haystack means more profit for the informed traders.

Proposition 6.1 (Informed-Trader Profit). *If the market maker uses the LASSO to identify and interpret feature-specific shocks and $N > N^*(Q, K)$, then the N informed traders have expected profits of*

$$\Pi(Q, \sigma_z) = C/2 \cdot \sqrt{K/N \cdot \log(Q)} \times \sigma_v \cdot \sigma_z \quad (29)$$

for some positive numerical constant $0 < C < \tilde{C}$ defined in Proposition 4.2.

What's interesting about the functional form of the informed traders' expected profits given in Proposition 6.1 is that the number of features, Q , and the volatility of noise trader demand, σ_z , enter multiplicatively. Feature-selection risk and noise trader demand risks are substitutes.

Substituting Risks. A natural follow up question is: What is the exchange rate between

feature-selection risk and noise trader demand risk? Suppose that you decreased noise trader demand volatility by a fraction Δ_{σ_z} :

$$Q \mapsto Q' = Q \cdot (1 + \Delta_Q) \quad (30a)$$

$$\sigma_z \mapsto \sigma'_z = \sigma_z \cdot (1 - \Delta_{\sigma_z}). \quad (30b)$$

At what rate would you have to add features to the market, Δ_Q , to leave the informed traders with exactly the same profit? It turns out that for small values of Δ_{σ_z} it is possible to answer this questions. I do this by expanding the expression for the informed traders' expected profit around any baseline level of (Q, σ_z) and solving for Δ_Q as a function of Δ_{σ_z} so that the first order terms cancel out:

$$0 = \frac{\partial}{\partial Q'} \Pi(Q', \sigma_z)|_{Q'=Q} \cdot \Delta_Q + \frac{\partial}{\partial \sigma'_z} \Pi(Q, \sigma'_z)|_{\sigma'_z=\sigma_z} \cdot \Delta_{\sigma_z}. \quad (31)$$

Corollary 6.1 (Substituting Risks). *Suppose you decreased the noise trader demand volatility by a fraction $\Delta_{\sigma_z} > 0$, then increasing the number of asset features by a fraction*

$$\Delta_Q = 2 \cdot \log(Q) \cdot \left(\frac{Q}{\sigma_z} \right) \times \Delta_{\sigma_z} \quad (32)$$

would leave informed trader expected profits and the price impact coefficient, λ , unchanged.

5.2. Seemingly Redundant Assets. In addition to being a substitute for noise-trader risk, feature-selection risk also makes novel predictions that are outside the scope of noise-trader risk. The textbook approach to learning about risks involves studying the prices of simple assets: 1 Arrow security for each shock. By contrast, compressed sensing theory asserts that an astute trader can identify feature-specific shocks from the prices of far fewer assets if a) the shocks are sparse and b) the chosen assets have extremely heterogeneous exposures to a large number of features.

Standard Approach. One way to learn about feature-specific shocks is to look at the price and demand of Arrow securities. For example, if there are Q payout relevant features:

$$\begin{bmatrix} d_1^{(A)} \\ d_2^{(A)} \\ d_3^{(A)} \\ \vdots \\ d_Q^{(A)} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}}_{\mathbf{X}^{(A)}} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_Q \end{bmatrix} + \frac{1}{\theta} \cdot \begin{bmatrix} z_1^{(A)} \\ z_2^{(A)} \\ z_3^{(A)} \\ \vdots \\ z_Q^{(A)} \end{bmatrix}. \quad (33)$$

This market setup is incredibly simple. The aggregate demand for the first Arrow security, $d_1^{(A)}$, tells the market maker if there has been a feature-specific shock to the first feature; the aggregate demand for the second Arrow security, $d_2^{(A)}$, tells the market maker if there has been a feature-specific shock to the second feature; the aggregate demand for the third

Arrow security, $d_3^{(A)}$, tells the market maker if there has been a feature-specific shock to the third feature; and so on.

Compressed-Sensing Intuition. Arrow securities are simple, but they are also wasteful. They don't exploit the fact that the market maker knows α is spiky and concentrated in only a few of its coordinates. Arrow securities are informative because they form an orthonormal basis. As a result, no 2 sets of feature-specific shocks can manifest themselves to the market maker in aggregate demand in exactly the same way. Yet, the market maker doesn't care about all possible collections of feature-specific shocks. He just cares about K -sparse shocks. Asset complexity gives a way for the market maker to exploit his knowledge of the sparsity of the feature-specific shocks.

For example, consider a collection of N derivative assets constructed by financial engineers out of the Q Arrow securities. These derivative assets will have an $(N \times Q)$ -dimensional exposure matrix $\mathbf{X}$,

$$\mathbf{X}_{N \times Q} = \mathbf{D}_{N \times Q}^{N \times Q} \mathbf{X}_{Q \times Q}^{(A)}, \quad (34)$$

where $\mathbf{D}$ is the matrix that governs how the Q Arrow securities are combined to create the N derivatives. Obviously, the N derivative assets can't have completely independent exposure to each of the Q payout-relevant features since $N \ll Q$. Some of the derivatives will have to have similar exposures to, say, crowded trade risk and S&P 500 inclusion risk. However, the market maker doesn't need the derivatives to be a completely linearly independent set of risk exposures. He just needs them to be sufficiently different.

Specifically, suppose that any $(2 \cdot K)$ columns of the $(N \times Q)$ -dimensional derivative feature-exposure matrix $\mathbf{X}$ are linearly independent. Then, any K -sparse signal $\alpha \in \mathbf{R}^Q$ can be reconstructed uniquely from $\mathbf{X}\alpha$. If not, then there would have to be a pair of K -sparse signals $\alpha, \alpha' \in \mathbf{R}^Q$ with $\mathbf{X}\alpha = \mathbf{X}\alpha'$; however, this would imply that $\mathbf{X}(\alpha - \alpha') = 0$ which is a contradiction. $\alpha - \alpha'$ is at most $(2 \cdot K)$ -sparse, and there can't be a linear dependence between $(2 \cdot K)$ columns of $\mathbf{X}$ by assumption. Thus, the market maker is happy to tolerate a little bit of redundancy. So long as traders can replicate the market's exposure to any $(2 \cdot K)$ features with fewer than N assets, aggregate demand shocks to the N assets will reveal which K feature-specific shocks have occurred.

Generalizing the Result. It is possible to generalize this result to random matrices. For an $(N \times Q)$ -dimensional matrix $\mathbf{X}$, the K -restricted isometry constant δ_K is the smallest number such that

$$\max_{|\mathcal{J}| \leq K} \|1/N \cdot \mathbf{X}_{[\mathcal{J}]}^\top \mathbf{X}_{[\mathcal{J}]} - \mathbf{I}\|_2 \leq \delta_K, \quad (35)$$

where $\mathbf{X}_{[\mathcal{J}]}$ denotes the columns of the matrix $\mathbf{X}$ corresponding to the J elements in the set $\mathcal{J}$. So, for example, if $\mathcal{J} = \{1, 7, 15\}$, then $\mathbf{X}_{[\mathcal{J}]}$ would be an $(N \times 3)$ -dimensional matrix

containing the first, seventh, and fifteenth columns of $\mathbf{X}$. For matrices with small restricted isometry constants, every subset of K or fewer columns is approximately an orthonormal system. Clearly, choosing $\mathbf{X}^{(A)} = \mathbf{I}$ via Arrow securities means that $\delta_K = 0$; however, [Candes and Tao \(2005\)](#) show that matrices with Gaussian entries, $x_{n,q} \stackrel{\text{iid}}{\sim} N(0,1)$, have small restricted isometry constants and allow for K -sparse recovery with very high probability whenever the number of measurements N is on the order of $N^*(Q,K) = K \cdot \log(Q/K)$. Proposition 6.2 characterizes the savings in required observations from examining complex derivative assets rather than Arrow securities.

Proposition 6.2 (Seemingly Redundant Assets). *If $N \geq N^*(Q,K)$, then a market maker using the LASSO with $\gamma = 2 \cdot (\sigma_z/\theta) \cdot \sqrt{2 \cdot \log(Q)}$ to study the aggregate demand for complex derivatives whose feature exposures are drawn $x_{n,q} \stackrel{\text{iid}}{\sim} N(0,1)$ can identify a K -sparse set of feature-specific shocks with probability greater than $1 - C_1 \cdot e^{-C_2 \cdot K}$ using*

$$K/Q \cdot \log(Q/K) \tag{36}$$

times fewer assets than a market maker studying the aggregate demand for Arrow securities on each of the Q features where $C_1, C_2 > 0$ are numerical constants.

RCT Analogy. There is an interesting analogy to randomized control trials here. That is, randomizing which assets get sold makes price changes and demand schedules more informative about feature-specific shocks in the same way that randomizing which subjects get treated in a medical study makes the experimental results more informative about the effectiveness of a drug. Why does randomization help? Suppose all of the people who got the real drug recovered and all of the people who got the placebo didn't. Randomly assigning patients to the treatment and control groups makes it exceptionally unlikely that the patients who took the real drug will happen to have some other trait, like a genetic variation, that actually explains their recovery. Randomizing feature exposures decreases the probability that 2 different K -sparse vectors α and α' are observationally equivalent when looking only at public market data.

6. EMPIRICAL EVIDENCE

Feature-selection risk should only matter when traders face a joint inference problem, that is, when traders have to simultaneously decide both which features are mispriced and how they should be correctly valued. Is there any evidence that traders actually care about this problem in the real world? Yes. Following the approach introduced in [Chinco, Clark-Joseph, and Ye \(2015\)](#), I show that using an estimation strategy which explicitly accounts for traders' joint inference problem increases the accuracy of out-of-sample return predictions at the monthly horizon by 144.3%, from $R^2 = 3.65\%$ to $R^2 = 9.35\%$! Thus, solving this joint inference problem is important to real-world traders.

6.1. Econometric Approach. Let's first look at what it means to say that an estimation strategy accounts for traders' joint inference problem.

Benchmark Regression. I begin by estimating a benchmark AR(1) specification using rolling 24-month sample periods,

$$rx_{n,t} = \hat{\phi}_0 + \hat{\phi}_1 \cdot rx_{n,t-1} + \epsilon_{n,t}, \quad (37)$$

like in [Jegadeesh \(1990\)](#). Formally, this amounts to minimizing the squared prediction error,

$$\hat{\phi} = \min_{\phi \in \mathbf{R}^2} \left\{ \frac{1}{24} \cdot \sum_{t=1}^{24} (rx_{n,t} - \{\phi_0 + \phi_1 \cdot rx_{n,t-1}\})^2 \right\}, \quad (38)$$

which is easy to do since there are many more observations, 24, than there are coefficients,

2. After fitting the regression coefficients, I then predict the subsequent month's returns:

$$E_t[rx_{n,t+1}] = f_{n,t}^{\text{AR}(1)} = \hat{\phi}_0 + \hat{\phi}_1 \cdot rx_{n,t}. \quad (39)$$

If $f_{n,t}^{\text{AR}(1)}$ is a good predictor of the realized excess return, $rx_{n,t+1}$, then traders only have to think about stock-specific considerations when predicting future returns. I measure the predictive power of $f_{n,t}^{\text{AR}(1)}$ by the R^2 of an out-of-sample regression,

$$rx_{n,t+1} = \tilde{a}_n + \tilde{b}_n \cdot \left(\frac{f_{n,t}^{\text{AR}(1)} - \mu_n^{\text{AR}(1)}}{\sigma_n^{\text{AR}(1)}} \right) + e_{n,t+1}, \quad (40)$$

where $\mu_n^{\text{AR}(1)}$ and $\sigma_n^{\text{AR}(1)}$ are the mean and standard deviation of the predictor $f_{n,t}^{\text{AR}(1)}$.

Penalized Regression. Next, I estimate the relationship between each stock's monthly excess returns and a large collection of Q predictors where $Q \gg 24$. Since there are more plausible predictors, Q , than months in the sample period, 24, the estimation strategy in Equation (38) is no longer valid. Instead, it's necessary to use a penalized regression. I use the least absolute shrinkage and selection operator (LASSO) as introduced in [Tibshirani \(1996\)](#) and used in Sections 4 and 5 above. This means choosing a $([Q+1] \times 1)$ -dimensional vector of coefficients using the optimization problem below:

$$\hat{\varphi} = \min_{\varphi \in \mathbf{R}^{Q+1}} \left\{ \frac{1}{2 \cdot 24} \cdot \sum_{t=1}^{24} \left(rx_{n,t} - \left\{ \varphi_0 + \sum_{q=1}^Q \varphi_q \cdot f_{q,t-1} \right\} \right)^2 + \lambda \cdot \sum_{q=1}^Q |\varphi_q| \right\}. \quad (41)$$

The LASSO penalty function, $\lambda \cdot \sum_{q=1}^Q |\varphi_q|$, sets all OLS coefficient estimates that are smaller than $|\varphi_q| < \lambda$ to zero as discussed in Section 4 above. Thus, the LASSO both selects and estimates the relevant coefficient loadings. This is the sense in which the LASSO explicitly incorporates traders' joint inference problem.

After fitting the regression coefficients, I again predict the subsequent month's returns:

$$E_t[rx_{n,t+1}] = f_{n,t}^{\text{LASSO}} = \hat{\varphi}_0 + \sum_{q=1}^Q \hat{\varphi}_q \cdot f_{q,t}. \quad (42)$$

If the predicted $f_{n,t}^{\text{LASSO}}$ is a good predictor of the realized excess return, $rx_{n,t+1}$, then traders' joint inference problem is crucial to predicting stock returns. I measure the predictive power of $f_{n,t}^{\text{LASSO}}$ by the R^2 of an out-of-sample regression,

$$rx_{n,t+1} = \tilde{a}_n + \tilde{c}_n \cdot \left(\frac{f_{n,t}^{\text{LASSO}} - \mu_n^{\text{LASSO}}}{\sigma_n^{\text{LASSO}}} \right) + e_{n,t+1}, \quad (43)$$

where μ_n^{LASSO} and σ_n^{LASSO} are the mean and standard deviation of the predictor $f_{n,t}^{\text{LASSO}}$.

Different Information. To make sure that both predictors aren't capturing the exact same information, I also run a regression with both predictors on the right-hand side:

$$rx_{n,t+1} = \tilde{a}_n + \tilde{b}_n \cdot \left(\frac{f_{n,t}^{\text{AR}(1)} - \mu_n^{\text{AR}(1)}}{\sigma_n^{\text{AR}(1)}} \right) + \tilde{c}_n \cdot \left(\frac{f_{n,t}^{\text{LASSO}} - \mu_n^{\text{LASSO}}}{\sigma_n^{\text{LASSO}}} \right) + e_{n,t+1}. \quad (44)$$

The logic here is simple. If the stock-specific information captured by the AR(1) model and feature-selection information captured by the LASSO model are really different kinds of information, then the R^2 of this combined regression will be roughly equal to the sum of the R^2 s from Equations (40) and (43).

6.2. Data Sources. I collect 79 different predictive variables at the monthly horizon from January 1990 to December 2010 from a variety of data sources. Monthly returns for NYSE stocks come from the [Wharton Research Data Service \(WRDS\)](#).

Ken French's Data Library. The bulk of the predictors come from Ken French's website. See http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html for more detailed variable definitions. I include factors representing the excess returns to the market portfolio as well as portfolios of small, large, growth, and value stocks as in [Fama and French \(1993\)](#). I also consider factors representing the excess returns to medium-term momentum ([Jegadeesh and Titman \(1993\)](#)) as well as to short- and long-term reversals ([Jegadeesh \(1990\)](#)). In addition, there are factors representing the excess returns to portfolios of high and low operating profit firms and high and low real-investment firms. Table 1 houses summary statistics for all of these predictors.

The same data library also contains data on the monthly excess returns to country- and industry-specific portfolios. I include factors for the following countries: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Switzerland, Sweden, and United Kingdom. Table 3 houses the summary statistics for these factors. The absence of many of the country-specific factors prior to January 1990 dictates the starting point for my sample period. Similarly, I include factors for 30 different SIC Code industries. See Table 4 for the relevant descriptive statistics.

Sentiment Variables. I incorporate data on a variety of market-sentiment indicators used

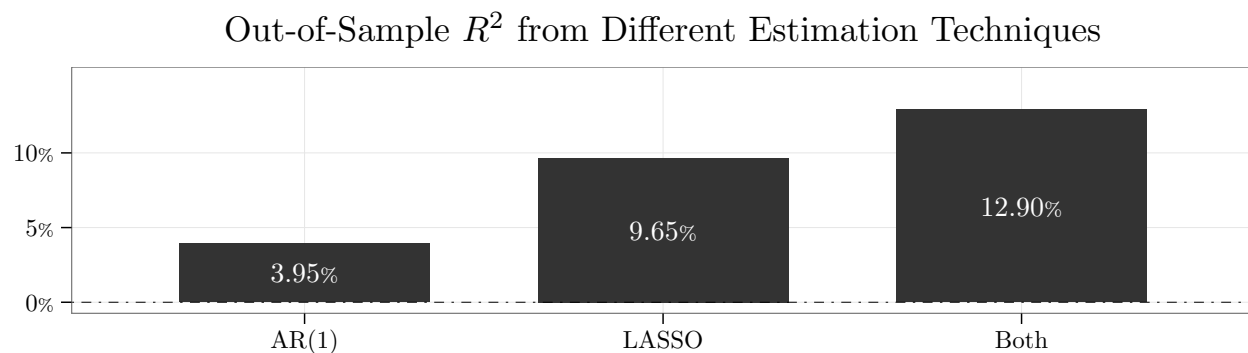


Figure 2. Average R^2 from an out-of-sample prediction of 1-month excess returns using an auto-regression, LASSO, or both. Data: Monthly returns for NYSE-listed stocks from January 1990 to December 2010. Reads: “Using the LASSO to predict returns boosts the out-of-sample R^2 by $(9.65 - 3.95)/3.95 \approx 144.3\%$.”

in Baker and Wurgler (2006). These data are all available on Jeff Wurgler’s website: <http://people.stern.nyu.edu/jwurgler/>. The dividend premium originally comes from Baker and Wurgler (2004), the number and first-day return on IPOs is defined in Ibbotson, Sindelar, and Ritter (1994), the average monthly turnover of NYSE stocks comes from the NYSE Factbook, the closed-end-fund discount is detailed in Neal and Wheatley (1998), and the equity share in new issues is originally outlined in Baker and Wurgler (2000). The sentiment index is a factor representing the first principal component of these six sentiment proxies over 1962-2005 time period, where each of the proxies has first been orthogonalized with respect to a set of macroeconomic conditions. Table 2 displays the relevant summary statistics.

Macroeconomic Variables. I also add a variety of other macroeconomic predictors: a recession indicator as defined by the National Bureau of Economic Research (NBER), factors representing the U.S. employment growth rate and the U.S. inflation rate from the Bureau of Economic Analysis (BEA), and a factor denoting the level of the VIX from the Chicago Board of Options Exchange (CBOE). Table 2 displays the summary statistics for these variables.

Additional Variables. Finally, there are three additional factors. The first two, a time-series momentum factor (Moskowitz, Ooi, and Pedersen (2012)) and a betting-against-beta factor (Frazzini and Pedersen (2014)), come from AQR’s data library. See <https://www.aqr.com/library/data-sets> for further details about their construction. The third factor is Pástor and Stambaugh (2003)’s liquidity-risk factor, which is available from Ľuboš Pástor’s website: http://faculty.chicagobooth.edu/lubos.pastor/research/liq_data_1962_2013.txt. See Table 1 for descriptive statistics.

6.3. Estimation Results. If real-world traders face a feature-selection problem—that is, if they have to both select and value the few relevant features when pricing each asset—then explicitly modeling this problem using the LASSO, just like the uninformed traders do in the model above, should improve out-of-sample return predictability for each stock. By contrast,

if traders don't face a feature-selection problem, then the LASSO shouldn't provide any additional out-of-sample predictive power. All the relevant information should be contained in the previous month's returns. The estimation results show that using the LASSO when making return predictions dramatically improves out-of-sample return predictability. Thus,

Summary Statistics for Market Factors

		Avg	Med	StD	Min	Max
Value-Weighted Market		0.5	1.1	4.5	-17.2	10.8
Small Stocks		1.1	1.7	6.2	-21.5	24.2
Large Stocks		0.8	1.2	4.4	-16.4	11.6
Growth Stocks		0.8	1.1	4.6	-15.4	14.2
Value Stocks		0.9	1.6	4.8	-22.1	16.6
Low Op. Profit Stocks		0.6	1.4	5.5	-21.4	12.5
High Op. Profit Stocks		0.9	1.3	4.2	-15.4	13.2
Low Investment Stocks		0.9	1.5	4.3	-16.1	10.5
High Investment Stocks		0.8	1.3	5.4	-18.5	13.6
Low E/P Stocks		0.8	1.1	4.7	-16.3	13.4
High E/P Stocks		1.1	1.6	4.6	-18.5	12.5
Momentum		0.6	0.8	5.3	-34.7	18.4
Short-Term Reversal		0.3	0.2	3.8	-14.5	16.2
Long-Term Reversal		0.4	0.3	2.6	-7.0	11.1
Time-Series Momentum		1.7	1.5	7.9	-17.4	24.4
Betting Against Beta		0.8	1.0	3.2	-10.5	10.7
Liquidity Factor		0.6	0.4	4.0	-10.1	21.0

Table 1. Monthly excess return on market factors from January 1990 to December 2010. Sources: Ken French's data library, AQR's data library, and Ľuboš Pástor's website. Reads: "The average value-weighted excess return on the market was 0.5% per month."

Summary Statistics for Sentiment and Macro Variables

		Avg	Med	StD	Min	Max
Sentiment Index		0.1	0	0.5	-0.9	2.5
Dividend Premium		-6.9	-7.2	9.8	-50.2	17.1
Number of IPOs		27.4	21.5	22	0	106
Return on IPOs		17.9	13.7	19.6	-19.9	116.2
Turnover		0.8	0.8	0.3	0.4	1.7
Closed-End-Fund Discount		6.1	6.2	4.5	-6	18.2
Equity Share in New Issues		0.3	2.1	57.9	-246.3	204.6
Recession Indicator		0.1	0	0.3	0	1
Employment Growth		0.1	0.1	0.2	-0.6	0.4
Inflation Rate		0.2	0.2	0.3	-1.9	1.2
VIX		20.4	19.2	8	10.8	62.6

Table 2. Monthly values for a variety of sentiment and macroeconomic variables from January 1990 to December 2010. Source: Jeffrey Wurgler's website, NBER, BEA, and CBOE. Reads: "The average level of the sentiment index was 0.1%."

this feature-selection problem appears very important for real-world traders.

LASSO Boosts Power. Figure 2 houses the key result: using an estimation strategy that accounts for traders' joint inference problem dramatically improves the accuracy of out-of-sample return predictions. If you just run a simple AR(1) model using rolling 24-month windows, you get an out-of-sample $R^2 = 3.95\%$; whereas, if you estimate the LASSO on the same 24-month windows, you get an out-of-sample $R^2 = 9.65\%$, an improvement of $(9.65 - 3.95)/3.95 \approx 144\%$. Moreover, these two estimation techniques are capturing fundamentally different information. When both the AR(1) and LASSO predictors are included in the same regression, the resulting R^2 is $12.9/(3.95 + 9.65) \approx 95\%$ of the sum of the R^2 's from the separate regressions.

Sparse Factor Loadings. The pattern of factor loadings that the LASSO chooses also supports the idea that traders face a feature-selection problem. Tables 5, 6, 7, and 8 reveal that, at most, only a small fraction of all NYSE stocks load each of the 79 different factors at any given time. What's more, there are many factors which are unhelpful for predicting future stock returns for months on end. For example, Table 5 shows that, while the excess returns to a momentum trading strategy predict the future returns of 18% of all NYSE stocks

Summary Statistics for Country-Specific Factors

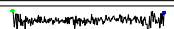
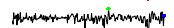
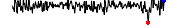
		Avg	Med	StD	Min	Max
Austria		0.8	1.0	7.0	-34.4	20.0
Australia		1.1	1.0	6.0	-27.7	17.3
Belgium		0.8	1.0	5.6	-30.8	16.7
Canada		1.0	1.4	5.6	-26.9	21.5
Denmark		1.0	1.4	5.7	-25.6	18.2
Finland		1.3	1.0	9.1	-29.0	30.8
France		0.8	1.2	5.9	-21.6	15.0
Germany		0.8	1.2	6.3	-23.3	22.0
Hongkong		1.4	1.2	7.6	-28.6	32.2
Italy		0.6	0.8	7.2	-23.7	21.2
Japan		0.1	0.0	6.4	-18.6	25.9
Netherlands		1.0	1.3	5.9	-30.1	17.1
New Zealand		0.6	0.9	6.4	-19.0	15.6
Norway		1.1	1.3	7.5	-31.0	19.5
Singapore		1.0	1.3	7.4	-28.3	29.3
Spain		0.9	1.0	6.7	-22.8	21.1
Sweden		1.2	1.5	7.6	-27.6	25.2
Switzerland		1.0	1.3	4.8	-14.7	14.8
United Kingdom		0.8	0.8	4.9	-20.4	14.3

Table 3. Monthly excess return on value-weighted country-specific portfolios from January 1990 to December 2010. Source: Ken French's data library. Reads: "The average value-weighted excess return on Australian stocks was 1.1% per month."

in January 2009, momentum is a significant predictor of zero NYSE stocks in March 1997.

More generally, the patterns of significance and insignificance for all of the factors appear spiky. None of the predictors is a useful indicator for all the stocks all the time. Factors suddenly lurch into importance and then shrink away. It just isn't obvious to traders ahead of time which factors they should be using to predict returns. This is exactly the sort of joint inference problem analyzed in the model above.

Summary Statistics for Industry-Specific Factors

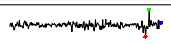
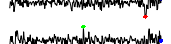
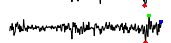
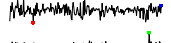
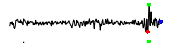
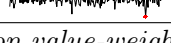
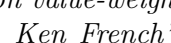
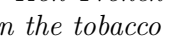
		Avg	Med	StD	Min	Max
Automobiles		0.8	1.0	8.0	-36.4	49.6
Beer		1.0	1.3	5.2	-19.8	16.4
Books		0.5	0.3	5.7	-26.5	33.1
Business Equipment		1.1	1.3	8.3	-31.8	25.1
Chemicals		0.9	1.2	5.7	-20.9	22.1
Clothes		1.0	1.6	6.7	-22.1	25.1
Coal		2.0	1.4	11.8	-37.9	44.0
Construction		0.8	1.2	6.1	-28.2	23.3
Electrical Equipment		1.3	1.1	6.5	-24.6	23.2
Fabricated Products		1.0	1.5	6.8	-29.9	20.8
Financials		0.9	1.3	5.8	-22.1	17.0
Food		0.9	1.1	4.2	-12.1	15.7
Gaming		1.0	1.3	7.1	-29.7	34.5
Healthcare		0.9	1.1	4.6	-12.3	16.5
Household		0.9	1.2	4.5	-14.3	18.5
Non-Auto Vehicles		1.2	1.8	6.2	-24.1	17.1
Non-Coal Mining		1.0	1.2	8.1	-34.5	35.6
Oil and Gas		1.0	0.7	5.3	-16.9	19.1
Other		0.3	0.6	6.0	-21.3	19.8
Paper		0.8	1.0	5.1	-18.5	21.0
Services		1.1	1.8	6.9	-19.3	23.8
Restaurants		0.9	1.3	5.2	-14.8	16.0
Retail		0.9	0.9	5.3	-14.6	14.3
Steel		0.9	0.9	8.6	-33.0	30.7
Tabacco		1.2	2.0	7.2	-24.9	32.5
Telecom		0.6	1.2	5.5	-16.2	21.3
Textiles		0.6	1.0	8.6	-28.5	59.0
Transportation		0.9	1.4	5.3	-16.7	14.5
Utilities		0.8	1.2	4.2	-12.7	11.7
Wholesale		0.7	1.3	4.8	-21.1	15.2

Table 4. Monthly excess return on value-weighted industry-specific portfolios from January 1990 to December 2010. Source: Ken French's data library. Reads: "The average value-weighted excess return on stocks in the tobacco industry was 1.2% per month."

7. RELATED LITERATURE

This paper borrows from and brings together several strands of literature.

Bounded Rationality. First, the current paper is closely related to the literature on bounded rationality; yet, there is a fundamental difference in approaches. Existing theories use cognitive constraints to induce boundedly rational decision making. For example, papers like Sims (2006) and Hong, Stein, and Yu (2007) suggest that cognitive costs force traders to use overly simplified mental models, and Gabaix (2013) derives the sort of mental models that traders would choose when facing ℓ_1 thinking costs. By contrast, I use bandwidth constraints on a market's *signals* rather than on a trader's *processing power* to generate similar behavior. Both channels are at work in asset markets. This paper is the first to articulate the bandwidth constraint on a finite set of market signals. To do this, I use the results from the compressed sensing literature, which originated with Candes and

Fraction of Stocks Loading on Each Market Factor

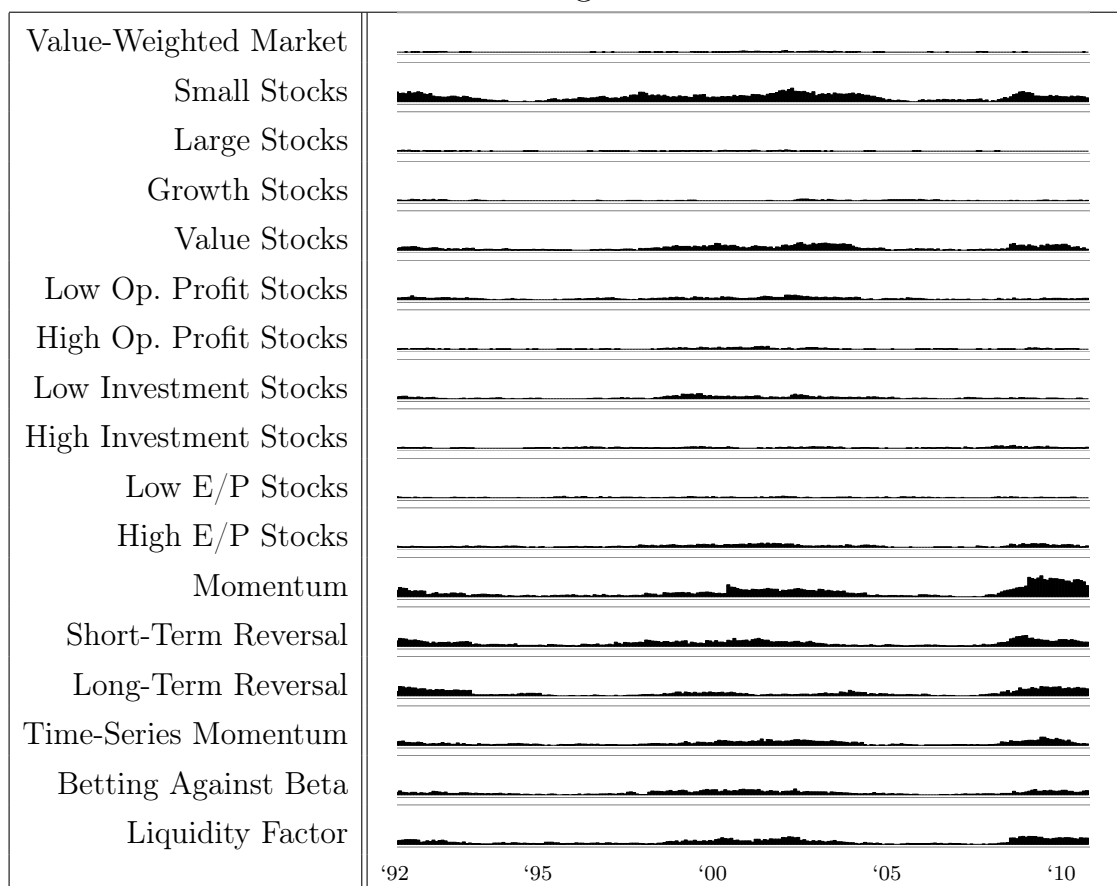


Table 5. Fraction of NYSE stocks each month that have non-zero loadings on each market factor when estimated using the LASSO in Equation (41). *y*-axis ranges from 0% to 25%. Reads: “The excess returns on the momentum portfolio in the previous month was a significant predictor for 18% of all NYSE stocks in January 2009.”

Tao (2005) and Donoho (2006).

Market Dimensions. Second, the model formulation relies on the fact that asset values are governed at least in part by a constantly changing cast of feature-specific shocks. Chinco (2015) provides evidence both that assets realize many different kinds of characteristic-specific shocks and also that it is hard for traders to identify which ones are relevant in real time. This assumption is consistent with, but separate from, existing asset-pricing models. On the theoretical side, it is possible to fit this high-dimensional problem into many popular asset-pricing models since they contain substantial amounts of theoretical “dark matter” in the language of Chen, Dou, and Kogan (2014).

On the empirical side, the high-dimensional and ever-changing nature of trader’s problem has been documented in a series of papers on data-snooping. For a representative sample, see Lo and MacKinlay (1990), Sullivan, Timmermann, and White (1999), and Kogan and Tian (2014). In particular, Kogan and Tian (2014) notes that parameter estimates for factor loadings are “highly sensitive to the sample period choice and the details of the factor construction. In particular, there is virtually no correlation between the relative model performance in the first and the second halves of the 1971-2011 sample period. Using a two-way sort on firm stock market capitalization (size) and characteristics to construct model return factors, an often used empirical procedure, similarly scrambles the relative model rankings.”

Fraction of Stocks Loading on Each Sentiment and Macro Variable

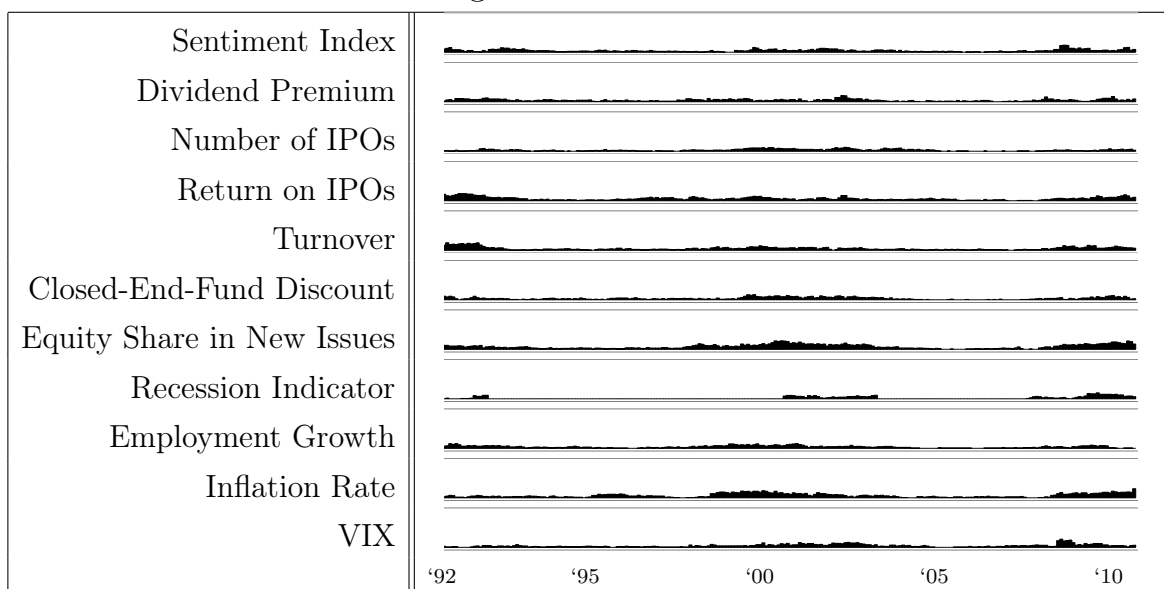


Table 6. Fraction of NYSE stocks each month that have non-zero loadings on each sentiment and macroeconomic variable when estimated using the LASSO in Equation (41). y-axis ranges from 0% to 25%. Reads: “The level of the VIX in the previous month was a significant predictor for 7% of all NYSE stocks in January 2009.”

Explanatory Power. Campbell, Lettau, Malkiel, and Xu (2001) also give evidence that the usual factor models only account for a fraction of firm-specific return volatility. For example, if you selected an NYSE/AMEX/NASDAQ stock at random in 1999, market and industry factors only accounted for 30% of the variation in its daily returns. Recent work by Ang, Hodrick, Xing, and Zhang (2006), Chen and Petkova (2012), and Herskovic, Kelly, Lustig, and Van Nieuwerburgh (2014) gives strong evidence that there is a lot of cross-sectional structure in the remaining 70% of so-called idiosyncratic volatility. That is, patterns in past idiosyncratic volatility are strong predictors of future returns. Thus, some portion of the 70% remainder appears to be neither permanent factor exposure nor fully idiosyncratic events.

Local Knowledge. Finally, this paper also gives a mathematical foundation for F.A. Hayek's notion of local knowledge. Indeed, Hayek (1945) gives trader who benefits from specialized

Fraction of Stocks Loading on Each Country-Specific Factor

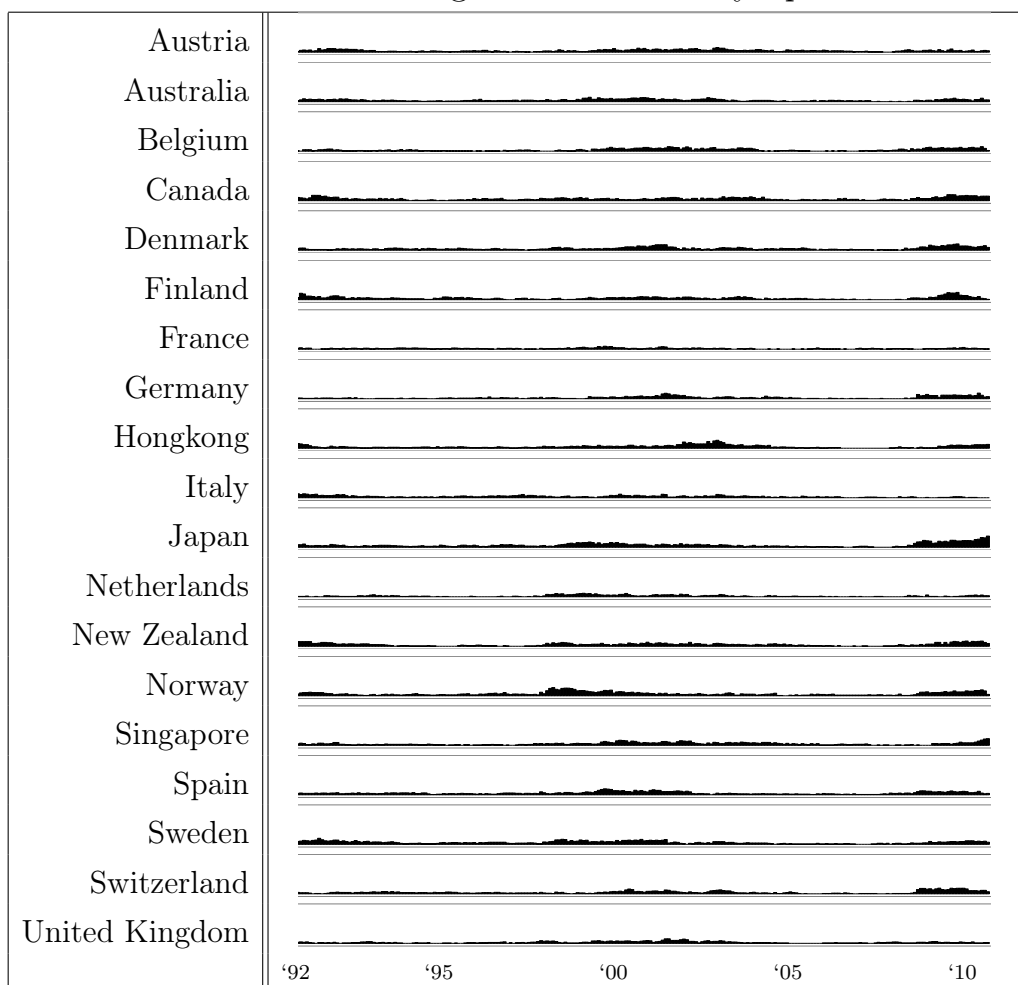


Table 7. Fraction of NYSE stocks each month that have non-zero loadings on each country-specific factor when estimated using the LASSO in Equation (41). *y*-axis ranges from 0% to 25%. Reads: “The excess returns on Norwegian stocks in the previous month was a significant predictor for 4% of all NYSE stocks in March 1999.”

Fraction of Stocks Loading on Each Industry-Specific Factor

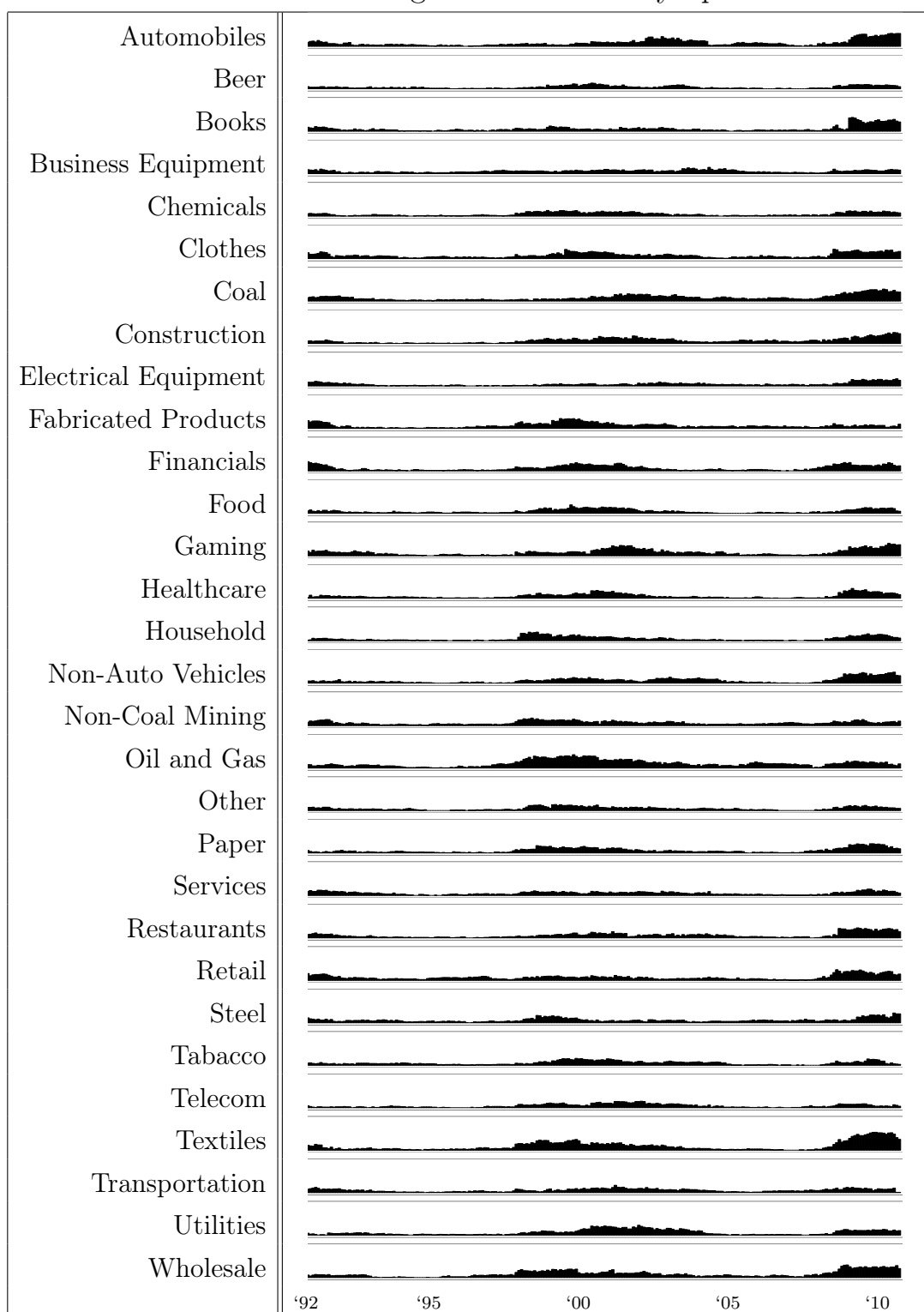


Table 8. Fraction of NYSE stocks each month that have non-zero loadings on each industry-specific factor when estimated using the LASSO in Equation (41). y-axis ranges from 0% to 25%. Reads: “The excess returns on steel stocks in the previous month was a significant predictor for 5% of all NYSE stocks in June 1999.”

experience with particular assets as a canonical example of a situation requiring local knowledge. One way to interpret the results is as something of an anti-Harsanyi doctrine and a microfoundation for the behavioral finance literatures on disagreement (see [Hong and Stein \(2007\)](#)) and noise trading (see [Black \(1986\)](#)). That is, this paper gives a situation where 2 rational Bayesian market makers can look at the exact same aggregate demand schedules for $N < N^*(Q, K)$ assets and not have the same posterior beliefs due to the dimensionality of the problem. I investigate these ideas further in Appendix [E](#).

8. CONCLUSION

Real-world traders have to simultaneously figure out both which asset features matter and also how much they matter. This paper develops the asset-pricing implications of traders' joint inference problem. Because traders have to simultaneously answer both 'Which features?' and 'How much do they matter?', the risk of selecting the wrong subset of features can spill over, warp their perception of asset values, and distort prices. Thus, feature-selection risk can limit market efficiency even though it stems from the inherent high-dimensional nature of modern asset markets and not some cognitive constraint or trading friction.

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APPENDIX A. PROOFS

Proof (Proposition 2.3). Each of the N asset-specific informed traders knows his own asset's true value, v_n , and solves the optimization problem below,

$$\max_{y_n} \mathbb{E}[(v_n - p_n) \cdot y_n \mid v_n],$$

giving the demand coefficient, $\theta(\lambda)$, up to the determination of λ :

$$y_n = \underbrace{\frac{1}{2 \cdot \lambda}}_{\theta(\lambda)} \cdot v_n.$$

I use the notation that $\mathbf{X}_{[\mathcal{K}]}$ denotes the measurement matrix $\mathbf{X}$ restricted to the columns $\mathcal{K}$ and that $\boldsymbol{\alpha}_{[\mathcal{K}]}$ denotes the coefficient vector $\boldsymbol{\alpha}$ restricted to the elements $\mathcal{K}$. Since an oracle has told the market maker which K features have realized a shock, he can use ordinary least squares to estimate $\boldsymbol{\alpha}$:

$$\hat{\boldsymbol{\alpha}}_{[\mathcal{K}],\text{OLS}} = \left\{ (\mathbf{X}_{[\mathcal{K}]}^\top \mathbf{X}_{[\mathcal{K}]})^{-1} \mathbf{X}_{[\mathcal{K}]}^\top \right\} \frac{\mathbf{d}}{\theta(\lambda)}.$$

Thus, the cross-section of aggregate demand gives the market maker a signal about each asset's fundamental value,

$$\hat{\mathbf{v}}_{\text{OLS}} = \mathbf{X}_{[\mathcal{K}]} \hat{\boldsymbol{\alpha}}_{[\mathcal{K}],\text{OLS}} = \frac{1}{\theta(\lambda)} \cdot \mathbf{d},$$

which has signal error:

$$\mathbb{E} \left[\frac{1}{N} \cdot \|\mathbf{v} - \hat{\mathbf{v}}_{\text{OLS}}\|_2^2 \right] = \frac{K}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2}.$$

Least squares prediction errors are normally distributed. In the limit as $N \rightarrow \infty$, the asset values are normally distributed since shocks, α_q , are bounded and selected independently from the same distribution. Using DeGroot (1969) updating to compute the market maker's posterior beliefs gives:

$$\text{Var}[v_n | \mathbf{d}] = \left(\frac{\frac{K}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2}}{\frac{K}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2} + \sigma_v^2} \right) \times \sigma_v^2 \quad \text{and} \quad \mathbb{E}[v_n | \mathbf{d}] = \underbrace{\frac{1}{\theta(\lambda)} \cdot \left(\frac{\sigma_v^2}{\frac{K}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2} + \sigma_v^2} \right)}_{\lambda} \cdot d_n.$$

Substituting in $\theta(\lambda) = 1/(2 \cdot \lambda)$ and simplifying gives the desired result. $\square$

Lemma A.1 (Fano's Error Inequality, Cover and Thomas (1991)). *Suppose x is a random variable with N outcomes $\{x_1, \dots, x_N\}$. Let y be a correlated random variable, $\text{Cor}[x, y] \neq 0$, and let $f(y)$ be the predicted value of x for some deterministic function $f(\cdot)$. Then, the following inequality holds,*

$$\Pr[x = f(y)] \geq 1 - \frac{M[x, y]}{\log_2(N)} - o(1),$$

where $M[x, y]$ denotes the mutual entropy between the random variables x and y .

Lemma A.2 (Mutual Information Bound, Cover and Thomas (1991)). *Suppose p is a random variable with N outcomes $\{p_1, \dots, p_N\}$ that represent probability distributions of $x \in \mathcal{X}$. Let $\hat{x} \in \mathcal{X}$ be a realization from 1 of the N probability distributions. Then, the following*

inequality holds,

$$M[p, y] \leq \frac{1}{N^2} \cdot \sum_{n, n'=1}^N \text{KL}[p_n(x|\hat{x}), p_{n'}(x|\hat{x})],$$

where $\text{KL}[p_n, p_{n'}]$ is the Kullback-Leibler divergence between the distributions p_n and $p_{n'}$.

Proof (Proposition 3.1). I show that if there exists some fixed constant C such that

$$N < C \cdot K_N \cdot \log(Q_N/K_N)$$

as $N \rightarrow \infty$, then there does not exist an inference rule $\phi \in \Phi$ such that $\text{FSE}[\phi] \rightarrow 0$.

The proof proceeds in 7 steps:

- (1) *Define variables.* Let $S = \binom{Q}{K}$ denote the number of feature subsets of size K and index each of these subsets with $\mathcal{K}_s$ for $s = 1, 2, \dots, S$. It is sufficient to consider the case where $\alpha_q = \alpha_{\min}$ for all $q \in \mathcal{K}_\star$ since this is easiest case. That is, if there is no selection rule ϕ that can identify the correct subset $\mathcal{K}$ when all of the coefficients are fixed at $\alpha_{\min}$, then there can be none when the coefficients are variable. Each subset is then associated with a distribution, p_s , given by

$$p_s = N(\alpha_{\min} \cdot \mathbf{X}[\mathcal{K}_s] \mathbf{1}, \mathbf{I})$$

for $s = 1, 2, \dots, S$ where $\mathbf{X}[\mathcal{K}_s]$ denotes the observed measurement matrix restricted to the columns $\mathcal{K}_s$, $\mathbf{1}$ denotes a $(K \times 1)$ -dimensional vector of 1s, and $\mathbf{I}$ denotes the $(K \times K)$ -dimensional identity matrix.

- (2) *Apply information inequalities.* Picking the right subset, $s \in \{1, \dots, S\}$, then amounts to picking the right generating distribution. Fano's inequality says that

$$\text{FSE}[\phi] = \Pr[\mathcal{K} = \phi(\mathbf{d}, \mathbf{X})] \geq 1 - \frac{M[p, \mathbf{d}|\mathbf{X}]}{\log_2(S)} - o(1).$$

I want to find conditions under which the right-hand side of this inequality is greater than 0. To do this, I need to characterize $M[p, \mathbf{d}|\mathbf{X}]$ which can be upper bounded:

$$M[p, \mathbf{d}|\mathbf{X}] \leq \frac{1}{S^2} \cdot \sum_{s, s'=1}^S \text{KL}[p_s(\mathbf{d}'|\mathbf{d}, \mathbf{X}), p_{s'}(\mathbf{d}'|\mathbf{d}, \mathbf{X})].$$

- (3) *Use functional form.* The optimal selection rule searches over all S feature subsets and tries to solve the program

$$\min_{s=1,2,\dots,S} \|\mathbf{d} - \alpha_{\min} \cdot \mathbf{X}[\mathcal{K}_s] \mathbf{1}\|_2^2 = \min_{s=1,2,\dots,S} \|\alpha_{\min} \cdot (\mathbf{X}[\mathcal{K}_\star] - \mathbf{X}[\mathcal{K}_s]) \mathbf{1} + \epsilon\|_2^2.$$

Plugging in the form of the optimization problem to characterize the Kullback-Leibler divergence and rearranging then gives

$$\begin{aligned} \text{FSE}[\phi] &= \Pr[\mathcal{K} = \phi(\mathbf{d}, \mathbf{X})] \\ &\geq 1 - \left(\frac{\frac{1}{2 \cdot S^2} \cdot \sum_{s, s'=1}^S \|\alpha_{\min} \cdot (\mathbf{X}[\mathcal{K}_s] - \mathbf{X}[\mathcal{K}_{s'}]) \mathbf{1}\|_2^2}{\log_2(S)} \right) - o(1). \end{aligned}$$

In order for $\text{FSE}[\phi] > 0$, it has to be the case that as $N \rightarrow \infty$ we have that

$$1 > \frac{1}{2 \cdot S^2} \cdot \frac{\sum_{s, s'=1}^S \|\alpha_{\min} \cdot (\mathbf{X}[\mathcal{K}_s] - \mathbf{X}[\mathcal{K}_{s'}]) \mathbf{1}\|_2^2}{\log_2(S)}.$$

- (4) *Characterize error distribution.* For any pair of subsets $(\mathcal{K}_s, \mathcal{K}_{s'})$ define the random variable as

$$h_{s,s'} = \|\alpha_{\min} \cdot (\mathbf{X}[\mathcal{K}_s] - \mathbf{X}[\mathcal{K}_{s'}]) \mathbf{1}\|_2^2.$$

Because assets have feature exposures, $x_{n,q} \stackrel{\text{iid}}{\sim} \mathcal{N}(0,1)$, $h_{s,s'}$ follows a χ_N^2 distribution,

$$h_{s,s'} \sim 2 \cdot \alpha_{\min}^2 \cdot (K - |\mathcal{K}_s \cap \mathcal{K}_{s'}|) \cdot \chi_N^2,$$

where $|\mathcal{K}_s \cap \mathcal{K}_{s'}|$ denotes the size of the set difference between the subsets $\mathcal{K}_s$ and $\mathcal{K}_{s'}$. For example, if there are $K = 4$ shocked features and $\mathcal{K}_s = \{1,2,5,9\}$ while $\mathcal{K}_{s'} = \{1,3,5,9\}$, then $|\mathcal{K}_s \cap \mathcal{K}_{s'}| = 1$.

- (5) *Bound mass in tail.* Using the tail bound for a χ_N^2 distribution, we see that

$$\Pr \left[\frac{1}{S^2} \cdot \sum_{s \neq s'} h_{s,s'} \geq 4 \cdot \alpha_{\min}^2 \cdot K \cdot N \right] \leq 1/2.$$

Thus, at least half of the S different subsets obey the bound below:

$$\frac{1}{2 \cdot S^2} \cdot \frac{\sum_{s,s'=1}^S \|\alpha_{\min} \cdot (\mathbf{X}[\mathcal{K}_s] - \mathbf{X}[\mathcal{K}_{s'}]) \mathbf{1}\|_2^2}{\log(S)} \leq \frac{4 \cdot \alpha_{\min}^2 \cdot K \cdot N}{\log_2(S)}.$$

- (6) *Formulate key inequality.* Thus, as long as

$$1 > \frac{4 \cdot \alpha_{\min}^2 \cdot K \cdot N}{\log_2(S)}$$

holds, the error rate will remain bounded away from 0 implying that

$$N > \left(\frac{1}{4 \cdot \alpha_{\min}^2 \cdot K} \right) \times \log_2(S)$$

is necessary for $\text{FSE}[\phi] \rightarrow 0$. The multiplier $(4 \cdot \alpha_{\min}^2 \cdot K)^{-1}$ is where the fixed constant C comes from in the result, so it is obvious that the constant will depend on the way that $\alpha_{\min}$ and K scale as the market grows large.

- (7) *Make cosmetic touch-up.* To make the formula above match, simply recall that:

$$S = \binom{Q}{K} \geq \left(\frac{Q}{K} \right)^K.$$

□

Lemma A.3 (Bound on Signal Error, [Candes and Plan \(2009\)](#)). *If $N \geq N^*(Q,K)$, then the LASSO estimate, $\hat{\alpha}_{\text{LASSO}}$, from the program in equation (22) using the tuning parameter $\gamma = 2 \cdot (\sigma_z/\theta) \cdot \sqrt{2 \cdot \log(Q)}$ obeys the inequality,*

$$\Pr \left[\frac{1}{N} \cdot \|\mathbf{X}\alpha - \mathbf{X}\hat{\alpha}\|_2^2 \leq \tilde{C}^2 \times \left(\frac{K \cdot \log(Q)}{N} \cdot \frac{\sigma_z^2}{\theta^2} \right) \right] \geq 1 - \frac{6}{Q^{2 \cdot \log 2}} - \frac{1}{Q \cdot \sqrt{2 \cdot \pi \cdot \log(Q)}},$$

with numerical constant $\tilde{C} = 4 \cdot (1 + \sqrt{2})$.

Proof (Proposition 4.2). Just as in Proposition 2.3, each of the N asset-specific informed traders knows his own asset's true value, v_n , and solves

$$\max_{y_n} \mathbb{E}[(v_n - p_n) \cdot y_n | v_n]$$

giving the demand coefficient, $\theta(\lambda)$, up to the determination of λ :

$$y_n = \underbrace{\frac{1}{2 \cdot \lambda}}_{\theta(\lambda)} \cdot v_n.$$

In the limit as $N \rightarrow \infty$, the asset values are normally distributed since shocks, α_q , are bounded and selected independently from the same distribution. However, now the cross-section of aggregate demand gives a signal about each asset's fundamental value with mean $\mathbf{v}$ and variance given in Lemma A.3.

Using DeGroot (1969) updating to compute the market maker's posterior beliefs gives:

$$\text{Var}[v_n|\mathbf{d}] = \left(\frac{C^2 \cdot \frac{K \cdot \log(Q)}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2}}{\sigma_v^2 + C^2 \cdot \frac{K \cdot \log(Q)}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2}} \right) \times \sigma_v^2 \quad \text{E}[v_n|\mathbf{d}] = \underbrace{\frac{1}{\theta} \cdot \left(\frac{\sigma_v^2}{\sigma_v^2 + C^2 \cdot \frac{K \cdot \log(Q)}{N} \cdot \frac{\sigma_z^2}{\theta(\lambda)^2}} \right)}_{\lambda} \cdot \mathbf{d}.$$

Noting that $\theta(\lambda) = 1/(2 \cdot \lambda)$ then gives the desired result after simplifying. $\square$

Proof (Proposition 6.1). Plugging the price impact and demand coefficients from Proposition 4.2 into the informed trader's optimization program in Equation 7 gives

$$\begin{aligned} \Pi(Q, \sigma_z) &= \text{E} \left(\max_{y_n} \text{E}[(v_n - \lambda \cdot \{y_n + z_n\}) \cdot y_n | v_n] \right) \\ &= \text{E}[(v_n - \lambda \cdot \{\theta \cdot v_n + z_n\}) \cdot \theta \cdot v_n] \\ &= \theta \cdot \frac{\sigma_v^2}{2}. \end{aligned}$$

Setting $\theta = C \cdot \sqrt{\log(Q)} \times \sqrt{K/N} \cdot (\sigma_z/\sigma_v)$ and simplifying gives the desired result. $\square$

Proof (Corollary 6.1). The functional form of the informed trader's expected profits comes from Proposition 6.1. Its partial derivative with respect to the number of features is

$$\frac{\partial}{\partial Q'} \Pi(Q', \sigma_z)|_{Q'=Q} = \frac{C}{2} \times \left(\frac{1}{2} \cdot \frac{1}{\sqrt{\frac{K}{N} \cdot \log(Q)}} \right) \times \left(\frac{K}{N \cdot Q} \right) \times \sigma_v \cdot \sigma_z.$$

Its partial derivative with respect to the amount of noise trader demand volatility is

$$\frac{\partial}{\partial \sigma'_z} \Pi(Q, \sigma'_z)|_{\sigma'_z=\sigma_z} = \frac{C}{2} \cdot \sqrt{\frac{K}{N} \cdot \log(Q)} \cdot \sigma_v.$$

In order for a tiny increase in the number of features, Δ_Q , to offset a tiny decrease in the amount of noise trader demand volatility, Δ_{σ_z} , the following condition has to hold:

$$\Delta_Q \times \frac{C}{2} \times \left(\frac{1}{2} \cdot \frac{1}{\sqrt{\frac{K}{N} \cdot \log(Q)}} \right) \times \left(\frac{K}{N \cdot Q} \right) \times \sigma_v \cdot \sigma_z = -\Delta_{\sigma_z} \times \frac{C}{2} \cdot \sqrt{\frac{K}{N} \cdot \log(Q)} \cdot \sigma_v.$$

Simplifying then yields the desired result. $\square$

Lemma A.4 (Bound on LASSO Recovery Error, Wainwright (2009b)). *If $N \geq N^*(Q, K)$, then the LASSO estimate, $\hat{\alpha}_{\text{LASSO}}$, from the program in equation (22) using the tuning parameter $\gamma = 2 \cdot (\sigma_z/\theta) \cdot \sqrt{2 \cdot \log(Q)}$ identifies the correct subset of feature-specific shocks*

with probability greater than

$$1 - C_1 \cdot \exp\{-C_2 \cdot K\}$$

for numerical constants $C_1, C_2 > 0$.

Proof (Proposition 6.2). First, consider the market maker studying Arrow securities that each have exposure to exactly 1 feature. The probability that any particular Arrow security will realize a shock is K/Q . The expected number of securities he needs to investigate before he sees all K feature-specific shocks is then given by the mean of negative binomial distribution with K failures:

$$\frac{(1 - K/Q) \cdot K}{K/Q} \asymp Q.$$

Second, consider the market maker studying complex derivatives. Lemma A.4 says that he can identify the correct features with exceedingly high probability using only $K \cdot \log(Q/K)$. The quotient gives the desired result. $\square$

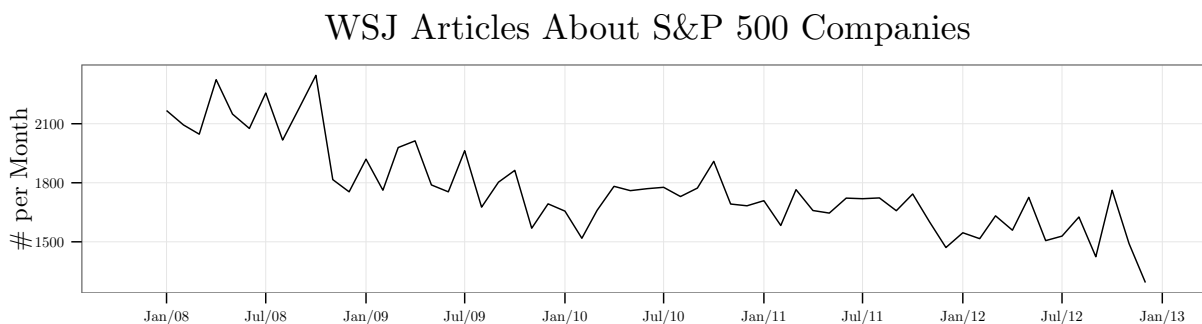


Figure 3. Number of Wall Street Journal articles about S&P 500 companies per month from January 2008 to December 2013. Reads: “There were roughly 1800 articles written about S&P 500 companies in the Wall Street Journal in July 2010.”

APPENDIX B. COUNTING ASSET FEATURES

A common question people have is: Is it possible to count the number of asset features, Q , in a market? Yes. I examine Wall Street Journal article keywords. The universe of keywords ever used is an estimate of the number of features. Even after controlling for the number of news articles, the number of asset features can vary by 2 orders of magnitude for S&P 500 stocks. The data are hand-collected from the [ProQuest newspaper archive](http://www.proquest.com/newspaper/archive).⁷ The resulting data set contains 106k articles over 5 years concerning 542 companies. Many articles reference multiple S&P 500 companies. Figure 3 plots the total number of articles in the database per month. There is a steady downward trend. The first part of the sample was the height of the financial crisis, so as markets have calmed down journalists have devoted fewer articles to corporate news relative to other things such as politics and sports.

Consistent with idea that companies get hit with new and different kinds of feature-specific shocks, Figure 4 shows that the vast majority of subject tags during the sample are only used in a couple of articles. While 50% of all subject tags are used in 3 or fewer articles, the

⁷See the online supporting materials at <http://www.alexchenco.com/ws-j-article-subject-tags/> for more details.

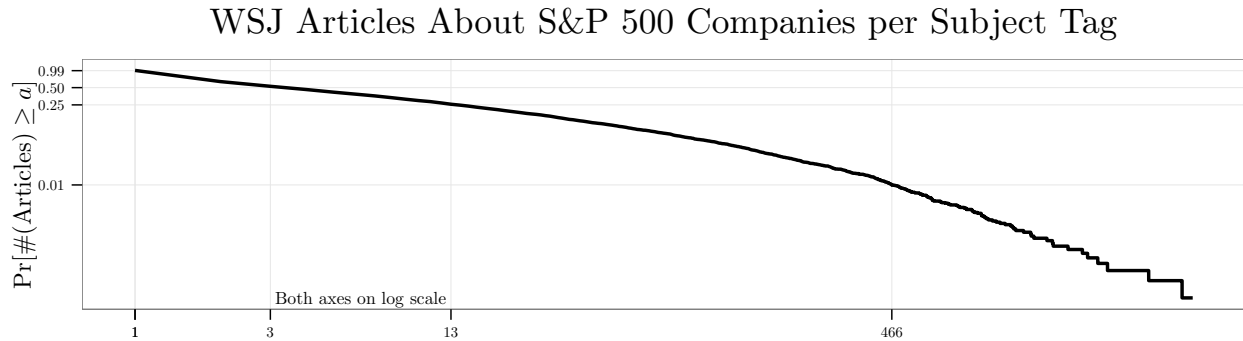


Figure 4. Number of Wall Street Journal articles per subject tag in articles about S&P 500 companies from January 2008 to December 2013. *x*-axis: Number of Wall Street Journal articles. *y*-axis: Fraction of all subject tags used in at least that many articles. Both axes are on a logarithmic scale. The break points on the *y*-axis define the 1%, 25%, 50%, and 99% quantiles. Reads: “While 50% of all subject tags are used in 3 or fewer articles, the most common 1% of the subject tags get used in 466 or more articles.”

most common 1% of the subject tags get used in 466 or more articles. Traders have to figure out which aspect of the company matters. This is clearly not an easy problem to solve. Lot’s of ideas are thrown around. Many of them must be either short lived or wrong. Roughly 1 out of every 4 topics worth discussing is only worth discussing once.

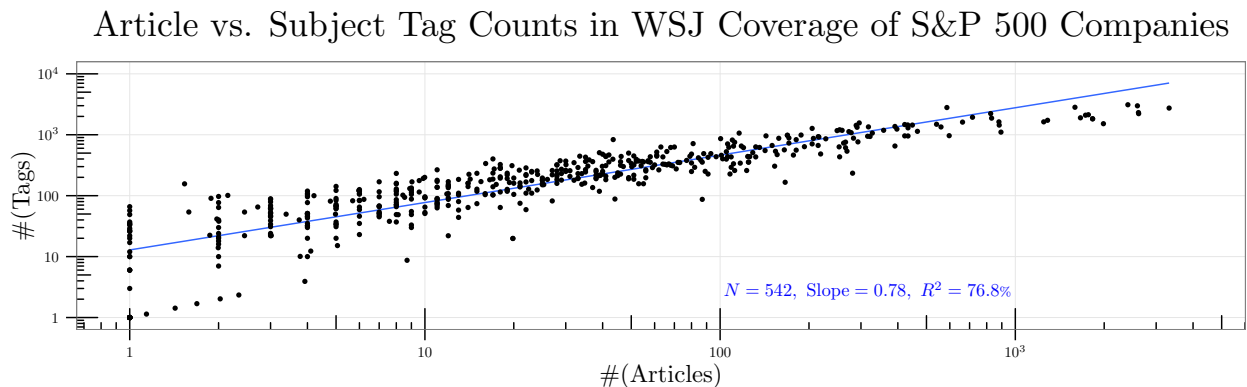


Figure 5. Number of Wall Street Journal articles about each S&P 500 company (*x*-axis) vs. number of unique subject tags used to describe each S&P 500 company (*y*-axis) over the period from January 2008 to December 2013. Both axes are on a logarithmic scale. Reads: “S&P 500 companies with between 100 and 200 articles in the Wall Street Journal typically have anywhere between 200 and 1000 distinct subject tags.”

In addition, I find that there is substantial heterogeneity in how many different topics people write about when discussing a company even after controlling for the number of total articles as shown in Figure 5. For example, there were 87 articles in the Wall Street Journal referencing Garmin (GRMN) and 81 articles referencing Sprint (S); however, while there were only 87 different subject tags used in the articles about Garmin, there were 716 different subject tags used in the articles about Sprint! This finding is consistent with the idea that some firms face a much wider array of shocks than others. In other words, the width of the market matters.

Evidence of Feature Selection Bound

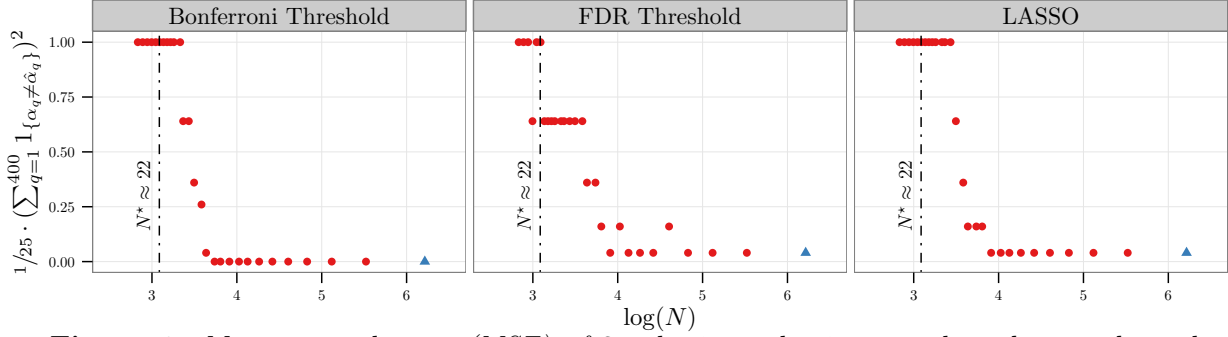


Figure 6. Mean squared error (MSE) of 3 selection rules in a market where each stock has $Q = 400$ features and the market realizes only $K = 5$ feature-specific shocks as the number of observations increases from $N = 15$ to $N = 400$. Left: Traders run univariate regressions and keep variables with t -stats exceeding $\sqrt{2 \cdot \log Q} \approx 3.46$. Middle: Traders use same regression procedure, but keep variables with p -values less than $0.25 \cdot (\hat{K}/Q)$ where $\hat{K}$ is the number of data-implied parameters in the model. Right: Traders select features using LASSO. Reads: “All 3 procedures display a sudden drop in MSE at $N^*(400, 5) \approx 22$.”

APPENDIX C. STANDARD INFERENCE PROBLEM

Traders in most information-based asset-pricing models solve a Gaussian inference problem as in [DeGroot \(1969\)](#). For example, market makers see N signals, d_n , which are informative about a fixed mean, $\bar{\alpha}$, that is contaminated with some noise, $\epsilon_n \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2)$. We might think about these signals as excess demand telling us about market sentiment à la [Baker and Wurgler \(2006\)](#),

$$d_n = \tilde{d}_n - E[\tilde{d}_n | \mathbf{f}] = \bar{\alpha} + \epsilon_n, \quad (45)$$

where $\tilde{d}_n$ denotes asset n 's gross demand, d_n denotes asset n 's excess demand, and $\mathbf{f}$ denotes a vector of factors. This framework has been extremely popular and productive because it leads to simple, intuitive, closed-form solutions. For example, if traders have prior beliefs, $\bar{\alpha} \stackrel{\text{iid}}{\sim} N(0, \sigma_\alpha^2)$, then their beliefs about $\bar{\alpha}$ after seeing N signals are given by

$$\text{Var}[\bar{\alpha} | \mathbf{d}] = \sigma_\alpha^2 \cdot \left(\frac{\sigma_\epsilon^2}{N \cdot \sigma_\alpha^2 + \sigma_\epsilon^2} \right) \quad \text{and} \quad E[\bar{\alpha} | \mathbf{d}] = \left(\frac{\sigma_\alpha^2}{N \cdot \sigma_\alpha^2 + \sigma_\epsilon^2} \right) \cdot \sum_{n=1}^N d_n. \quad (46)$$

See [Veldkamp \(2011\)](#) for an excellent overview of this literature.

APPENDIX D. NUMERICAL EXAMPLE

Suppose that stocks have $Q = 400 \gg 7$ features and the market realizes $K = 5 > 1$ feature-specific shocks so that aggregate demand is given by

$$d_n = \tilde{d}_n - E[\tilde{d}_n | \mathbf{f}] = \sum_{q=1}^{400} \alpha_q \cdot x_{n,q} + \epsilon_n \quad \text{and} \quad 5 = \|\boldsymbol{\alpha}\|_{\ell_0} = \sum_{q=1}^{400} 1_{\{\alpha_q \neq 0\}}. \quad (47)$$

Here, $x_{n,q} \stackrel{\text{iid}}{\sim} N(0, 1)$ denotes stock n 's exposure to the q th feature, $\epsilon_n \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2)$ denotes idiosyncratic noise for stock n , and $\alpha_q = 1/\sqrt{K}$ for all $q \in \{q' \in \mathcal{Q} : \alpha_{q'} \neq 0\}$. Notice that in this extension, I am no longer hand-picking each stock's feature exposures.

There are a number of statistical techniques to identify which 5 of the 400 features have realized a shock. First, you might try forward stepwise regression as in [Weisberg \(2005\)](#),

$$d_n = \hat{\alpha}_q \cdot x_{n,q} + \varsigma_n, \quad (48)$$

for all $q = 1, 2, \dots, Q$ keeping only the variables whose t -statistics exceed the Bonferroni threshold of $\sqrt{2 \cdot \log Q} \approx 3.46$. The left panel of Figure 6 shows the mean squared error from this approach. As you would expect, if there are more observations for you to analyze—that is, moving left to right, then you are better able to identify the 5 shocked features. However, the change doesn't happen gradually. Your error rate in interpreting aggregate demand schedules suddenly plummets once you've seen $N^*(400, 5) \approx 22$ observations. This is the feature-selection bound.

Here is the interesting part. This critical number is independent of the statistical procedure you use. For example, the middle panel shows the results if you were to use the same stepwise-regression procedure but keep only the variables whose p -values were less than $0.25 \cdot (\hat{K}/Q)$, with $\hat{K}$ denoting the total number of data-implied parameters in the model. This cutoff is known as the false-discovery-rate (FDR) threshold and comes from [Donoho and Jin \(2006\)](#). Alternatively, the right panel shows the results if you were to use the least absolute-shrinkage and selection operator (LASSO) as in [Tibshirani \(1996\)](#). Each panel displays a sudden drop in the error rate just after the feature-selection bound has been reached. The bound is a generic property of the high-dimensional inference problem.

APPENDIX E. LOCAL KNOWLEDGE

Entropy in Shocks ($\mathcal{K}$) vs. Entropy in Measurements ($\mathbf{X}$)

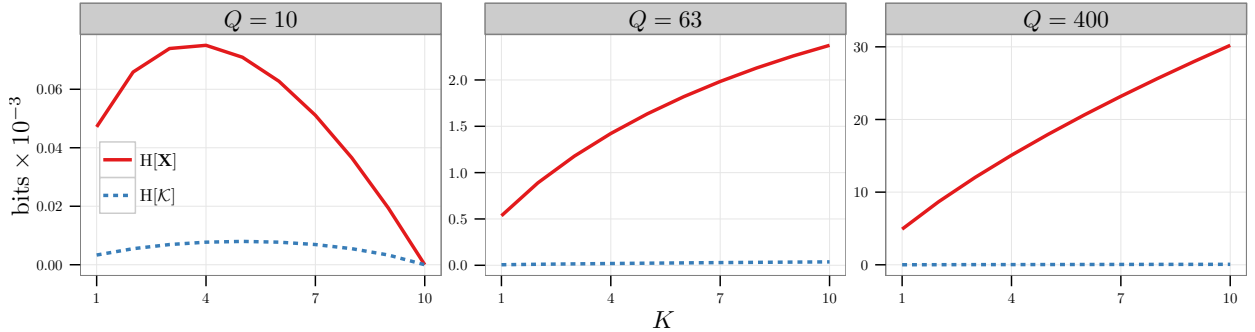


Figure 7. Entropy needed to transmit both the choice of K feature-specific shocks, $H[\mathcal{K}]$ (blue, dashed), and the feature-exposure matrix for the N^* observations needed to identify them, $H[\mathbf{X}]$ (red, solid), as the number of shocks grows from $K = 1$ to $K = 10$ for $Q \in \{10, 63, 400\}$. Reads: “ $H[\mathbf{X}] = 18 \times 10^3$ bits and $H[\mathcal{K}] = 36$ bits when $Q = 400$ and $K = 5$ corresponding to a vertical line through the right panel at $K = 5$. Thus, it takes 500 times as much information to measure all of the feature exposures for the $N^* \approx 22$ assets needed to identify $\mathcal{K}$ as it does to record the actual configuration entropy of $\mathcal{K}$.”

I conclude this paper by examining the role of local knowledge in this analysis. The goal in this subsection is to shed light on how local knowledge differs from the usual notions of cognitive costs in the economics and finance literature. For example, in existing information-based asset-pricing models, the cost of a signal typically scales with how much smarter it makes you as measured by an increase in the precision of your posterior beliefs or a reduction

in their entropy. See [Veldkamp \(2006\)](#) for a representative example. By contrast, in the current paper the cost of acquiring knowledge about which K features have realized a shock scales with the number of measurements necessary to uncover this information. What's more, the entropy bound up in these measurements typically exceeds the actual entropy of the signal by an order of magnitude or more. I call this gap the amount of local knowledge in the market.

To make these statements more precise, I first calculate how much information it would take to convey which K feature-specific shocks have occurred in a market with Q characteristics and K shocks. Let $\binom{Q}{K} = W$ denote the number of ways to select K characteristics from among Q possibilities. The amount of information in the signal is then given by the configuration entropy of $\mathcal{K}$ in units of bits:

$$H[\mathcal{K}] = - \sum_{w=1}^W \frac{1}{W \cdot \log(2)} \cdot \log \left(\frac{1}{W} \right). \quad (49)$$

Yet, in order for a market maker to uncover this signal by studying the cross-section of aggregate demand, he has to observe the feature exposures of $N^*(Q, K)$ assets. This is a $(N^* \times Q)$ -dimensional matrix with elements $x_{n,q} \stackrel{\text{iid}}{\sim} N(0,1)$. Each element in this matrix is a single measurement. Thus, the amount of information necessary to store all of these measurements in units of bits is given by

$$H[\mathbf{X}] = - \frac{1}{2 \cdot \log(2)} \cdot \log \left[\frac{1}{(2 \cdot \pi \cdot e)^{Q \cdot N^*}} \right]. \quad (50)$$

The proposition below characterizes how the value to market-wide arbitrageurs of immediately discovering the local knowledge scales with the signal recovery bound.

Proposition D (Local Knowledge). *The entropy of the measurements needed to discover which K feature-specific shocks have occurred exceeds the configuration entropy of the shocks:*

$$\text{Local Knowledge} = H[\mathbf{X}] - H[\mathcal{K}] \geq 0. \quad (51)$$

Proof. $H[\mathcal{K}]$ is maximized with shock probability for each of the Q features is $1/2$. The entropy of Q independent random normal variables is at least as large as that of a binomial distribution with Q draws and probability $1/2$. $N^*(Q, K) \geq 1$. $\square$

In order to uncover which K feature-specific shocks have occurred, a market maker has to observe the prices of a bunch of assets with randomly assigned feature exposures so that it is really unlikely that 2 different sets of K -sparse shocks can produce the same pattern in aggregate demand by pure chance. This means that the feature-exposure matrix, $\mathbf{X}$, is a big, random, unstructured matrix. As a result, it takes a lot of entropy to store it. For instance, when $Q = 400$ and $K = 5$, it takes 500 times as much information to measure all of the feature exposures for the $N^* \approx 22$ assets needed to identify $\mathcal{K}$ as it does to record the actual configuration entropy of $\mathcal{K}$ as shown in Figure 7.